CASE STUDY

Russia-Attributed Threat Actors: The Public Record

A TeamStarWolf case study. Published 3 November 2026; every source checked as of that date.

Executive summary

Anyone who has defended a network in Europe or North America over the past decade has met the activity in this study, usually under a nickname. Fancy Bear in an election inbox. Cozy Bear in a vaccine lab. Sandworm switching off substations in Ukraine. Snake infrastructure in more than 50 countries. Star Blizzard writing to a think-tank researcher as a trusted colleague. Behind those nicknames, governments have now named the paying customer: three military intelligence units of the GRU, the SVR foreign intelligence service, two centers of the FSB security service and a Ministry of Defense research institute. This study gathers the public record on those units and the groups the community tracks for them, dates every event, and grades every claim by the source that supports it.

The record supports five plain statements. First, attribution is unusually specific. Joint advisories from the US, the UK and their partners map tracked groups to named units: GRU Unit 26165 to APT28, GRU Unit 74455 to Sandworm, GRU Unit 29155 to the WhisperGate operators, the SVR to APT29, FSB Center 16 to Berserk Bear and the Snake toolset, FSB Center 18 to Star Blizzard, and TsNIIKhM to the Triton operators (**Confirmed**, AA22-110A, AA23-129A, AA23-341A, AA24-249A [30] [38] [40] [51]). Second, the court record is deep. US grand juries indicted GRU officers in 2018 (twice) and 2020, four Russian government employees over energy-sector intrusions in 2022, an FSB officer in 2023 and five Unit 29155 officers in 2024 (**Confirmed**, FBI wanted notices [7] [9] [14] [29] [39] [52]). Third, the way in is ordinary. Every one of the 26 CVEs named in five joint advisories and an FBI notice on these actors is in CISA's Known Exploited Vulnerabilities catalog, and FSB Center 16 was still exploiting a 2018 Cisco flaw in 2025 (**Confirmed** [92] [56]). Fourth, the services overlap and the labels move. The December 2025 attack on Poland's energy sector was assigned to Sandworm by ESET at medium confidence, while CERT Polska found a high degree of infrastructure overlap with the cluster known as Static Tundra, Berserk Bear, Ghost Blizzard or Dragonfly (which the FBI attributes to FSB Center 16 [56]), before the UK and EU formally attributed it to FSB Centre 16 in July 2026 (**Confirmed** [79] [58] [61]). Fifth, healthcare sits in the record in three ways: espionage against vaccine research (APT29

and APT28 in 2020), collateral damage from destructive malware (NotPetya in 2017), and sector lists in broad targeting advisories (Unit 29155 in 2024, FSB Center 16 in 2026). Disruptive attacks on hospitals themselves come from Russia-based criminals and pro-Russia hacktivists, not from a named state unit (**Inferred** from [12] [50] [32] [51] [62] [33] [34]).

What this means for a security program is practical. The most-stated technique across these advisories is exploitation of a public-facing application, so internet-facing patching against KEV is the shared first control against all three services. Identity is the second front: password spraying, dormant accounts, stolen tokens, MFA fatigue and session-cookie theft recur in the SVR, GRU and FSB advisories alike. End-of-life routers and switches are exposure in their own right. Healthcare organisations should plan for state espionage against research and for criminal disruption of care, and should not assume that only one of the two applies.

> Evidence grade: the government, court and ATT&CK record is **Confirmed** against primary sources opened on 3 November 2026. Vendor attributions are reported at the confidence each vendor states. Two press items (the Dutch Laundry Bear attribution and the Merck NotPetya ruling), the ATT&CK descriptions of the 2016 election intrusions, the indictments, TEMP.Veles, ALLANITE and the criminal groups, the Ukrainian attributions of Gamaredon and Kyivstar, and HC3's SolarWinds hospital claim are **Reported**, because only a secondary or summary source was opened for them.

Key facts at a glance

Field	Value
Services named by governments	GRU: 85th GTsSS (Unit 26165), GTsST (Unit 74455), 161st Specialist Training Center (Unit 29155); SVR; FSB: Center 16 (Unit 71330) and Center 18 (Unit 64829); Ministry of Defense research institute TsNIIKhM [30] [51]
Main tracked groups	APT28 G0007, Sandworm Team G0034, Ember Bear G1003, APT29 G0016, Turla G0010, Dragonfly G0035, Star Blizzard G1033, Gamaredon Group G0047, TEMP.Veles G0088
Earliest dated activity	Snake development from late 2003 (FSB, per nine agencies) [38]; APT28 and Turla active since at least 2004 (ATT&CK) [80] [83]
US indictments	2018-07-13 (12 GRU officers), 2018-10-03 (7 GRU officers), 2020-10-15 (6 GRU officers), announced 2022-03-24 (4 Russian government employees), 2023-12-05 (an FSB officer and a Callisto Group affiliate), announced 2024-09-05 (5 Unit 29155 officers and a civilian) [7] [9] [14] [29] [39] [52]
Sanctions	US Treasury: TsNIIKhM (2020-10-23), six technology companies serving the intelligence services (2021-04-15); UK: two Star Blizzard members, Peretyatko and Korinets (2023-12-07); UK: 24 designations announced with the first joint UK-EU cyber sanctions package (2026-07-13) [17] [20] [41] [61]
CVEs in the advisories	26 named in AA20-296A, AA22-011A, AA23-347A, AA24-249A, AA25-141A and FBI PSA I-082025-PSA; all 26 in KEV; 11 marked for known ransomware use [92]
ATT&CK v20.0 footprint	330 Enterprise techniques across nine groups, eight campaigns and eight advisories (252 at group level, 41 campaign-only, 37 advisory-only) [100]
Healthcare in the record	SVR vaccine-research espionage (2020); NotPetya collateral (2017); Unit 29155 (2024) and FSB Center 16 (2026) sector lists; hospital disruption by KillNet and Russia-based ransomware crews [12] [32] [51] [62] [35] [34]
Latest public record	UK-EU formal attribution of the December 2025 Poland energy attack to FSB Centre 16 and a 19-agency router advisory, both 2026-07-13 [61] [62]; FBI notice on APT28 SOHO router compromise, 2026-04-07 [59]

Key findings

- **Governments name units, not just countries.** AA22-110A, issued by five governments on 20 April 2022, lists FSB Center 16 (unit 71330), FSB Center 18 (unit 64829), the SVR, GRU 85th GTsSS (unit 26165), GRU GTsST (unit 74455) and TsNIIKhM as the Russian organizations that run cyber operations, and maps them to Berserk Bear, APT29, APT28, Sandworm and TEMP.Veles (**Confirmed** [30]). AA24-249A adds GRU Unit 29155, separate from the other two GRU units (**Confirmed** [51]).
- **Confidence has firmed over time.** FireEye called APT28 "most likely sponsored by the Russian government" in 2014; the NCSC and partners said "almost certainly" Unit 26165 in 2023. Dragonfly went from "Russian government cyber actors" via Symantec (2018) to "almost certainly" FSB Center 16 (2022) to a formal UK-EU attribution (2026) (**Confirmed** for each statement [65] [37] [5] [30] [61]; the trend is **Inferred**).
- **The court record reaches the GRU, the FSB and TsNIIKhM; no SVR officer is charged in the record opened.** The 2018 election indictment charged officers of both Unit 26165 and Unit 74455; the 2020 indictment charged Unit 74455 officers for Ukraine, NotPetya, Olympic Destroyer and Georgia; the 2022

case charged a TsNIIKhM employee over safety-system attacks; the 2023 case charged an FSB officer; the 2024 case charged Unit 29155 officers over WhisperGate (**Confirmed** [7] [8] [14] [29] [39] [52]).

- **Old, known flaws carry the intrusions.** All 26 CVEs named in the five joint advisories and the FBI notice are in KEV; the earliest entries date from the catalog's launch on 3 November 2021. CVE-2018-0171 in Cisco Smart Install has been in KEV since that day, yet the FBI reported FSB Center 16 exploiting it in August 2025 (**Confirmed** [92] [56]; the patch-debt reading is **Inferred**).
- **Identity attacks cross every service.** Password spraying is stated for the SVR (AA24-057A), the GRU (AA24-249A, AA25-141A) and earlier Russian state actors (AA22-011A); Star Blizzard steals session cookies that defeat ordinary MFA (AA23-341A); Midnight Blizzard sprayed a legacy Microsoft test tenant without MFA in late 2023 (**Confirmed** [45] [51] [55] [26] [40] [69]).
- **Network devices are a long campaign.** TA18-106A (2018) warned of Russian state exploitation of routers through Smart Install, SNMP and GRE; APT28 used a 2017 Cisco flaw for Jaguar Tooth in 2021; APT28 used compromised EdgeRouters to proxy traffic and harvest credentials (2024) and hijacked DHCP and DNS on compromised SOHO routers for adversary-in-the-middle attacks (FBI, 2026); FSB Center 16 copied router configurations worldwide through 2026 (**Confirmed** [6] [37] [46] [59] [62]).
- **Sources disagree on the newest attacks.** ESET assigns the 29 December 2025 Poland wiper attempt to Sandworm at medium confidence; CERT Polska reports a high degree of infrastructure overlap with the Static Tundra, Berserk Bear, Ghost Blizzard and Dragonfly cluster and does not mention Sandworm; ATT&CK records both readings in campaign C0063 and associates it with no group (**Confirmed** [79] [58] [90]). The operational record of the attack and the attribution dispute are covered in ICS attacks and advisories, 2024 to 2026.
- **Healthcare is a target of espionage and a casualty of disruption.** APT29 went after COVID-19 vaccine research in Canada, the US and the UK in 2020; SVR TeamCity exploitation in 2023 reached a medical-device software maker; HC3 recorded KillNet's 28 January 2023 DDoS wave on hospitals; the FBI counted at least 16 Conti attacks on US healthcare and first-responder networks in a year (**Confirmed** [12] [42] [35] [23]).

Top recommendations

1. Patch internet-facing systems against KEV first, then against every CVE these advisories name. `T1190` Exploit Public-Facing Application is stated in six of the eight advisories with ATT&CK tables opened for this study, across GRU, SVR and FSB units (**Inferred** from [16] [26] [38] [42] [51] [55]). Treat "in KEV and named in a Russia advisory" as the top tier of your remediation SLA.
2. Inventory and retire end-of-life network devices. Disable Cisco Smart Install, SNMPv1 and v2c and Telnet, block TCP 4786 at the boundary and keep management planes off the internet, as TA18-106A asked in 2018; the FBI's 2025 notice shows the same Smart Install flaw still being exploited on end-of-life devices (**Confirmed** [6] [56]).
3. Move to phishing-resistant MFA and close the identity gaps the advisories list: dormant accounts, over-privileged service accounts, legacy OAuth applications with app-only mail rights, open device registration (**Confirmed** [45] [51] [55] [69]). Use Multi-factor Authentication and Credential Hardening as the countermeasure pages.
4. Block outbound SMB and NTLM to the internet and plan NTLM retirement. AA25-141A ties this to the Outlook NTLM flaw CVE-2023-23397 that Unit 26165 used (**Confirmed** [55]).

5. Put EDR on mail servers and domain controllers first, alert when EDR goes silent, and keep memory forensics available for kernel implants such as Snake (**Confirmed** [55] [42] [38]).
6. Healthcare and life-science organisations: treat research data as an espionage target and clinical operations as a disruption target. Use the NCSC 2020 vaccine advisory and AA24-249A sector lists for the first, and the HC3 KillNet notes and ransomware advisories for the second (**Inferred** from [12] [51] [35] [18]).
7. Test controls against the study layer. AA24-249A and AA23-347A ask defenders to validate security controls against the ATT&CK techniques they list (**Confirmed** [51] [42]); Part 8 gives the layer.

Scope and method

This study covers activity that a government, a court record, MITRE ATT&CK or a named vendor attributes to a Russian state organization, from the earliest dated record (FSB development of Snake from late 2003) to the UK-EU attribution and the 19-agency router advisory of 13 July 2026. It adds a short, separately labelled section on Russia-nexus groups that no government opened for this study attributes to a named service, and on Russia-based criminal and hacktivist groups, because both matter to defenders and especially to healthcare. It draws on six kinds of source: US court records restated in FBI wanted notices; joint government advisories and attribution statements from the US, UK, France, Czechia and others; sanctions designations; vendor threat reports; the MITRE ATT&CK, D3FEND and CAPEC record; and the TeamStarWolf library's own data files at commit `e85b5594`. Every source in the Sources table was opened on 3 November 2026.

Each claim carries one of four grades.

Grade	Meaning
Confirmed	The primary source that published the fact (the court, agency, vendor, victim or MITRE page) was opened on 3 November 2026 and says it. For an attribution, Confirmed means the source makes that attribution, at the confidence it states; it does not upgrade the source's own confidence
Reported	Only a secondary source (press coverage, or ATT&CK's summary of an indictment or report that was not itself opened) was available
Demonstrated	A public reproduction exists. No claim in this study rests on one
Inferred	This study's own reasoning from Confirmed facts, labelled where it appears

This study grades by source type (primary opened versus secondary only), the scheme the newer case studies use; under the README's original scheme, single-vendor reports would read as Reported.

Four rules govern the names and the boundaries.

1. The study says which name each source uses ("Microsoft's Midnight Blizzard", "Mandiant's APT44", "CERT Polska's Static Tundra cluster") rather than merging labels, and a source's own hedge travels with its claim ("almost certainly", "medium confidence", "highly likely").
2. Where a government names a unit and ATT&CK does not, both are reported. ATT&CK's Star Blizzard page, for example, names no service; the joint advisory it cites says FSB Centre 18 (**Confirmed** [87] [40]).
3. Events that have a home study elsewhere in the library are summarized here in one paragraph and linked. Sandworm's operations (the Ukraine grid attacks, NotPetya, Olympic Destroyer, Industroyer2) belong in

SANDWORM_OPERATIONS.md; the SolarWinds Orion compromise belongs in SOLARWINDS_SUNBURST.md; the ALPHV attack on Change Healthcare belongs in Change Healthcare 2024; the operational record of the 29 December 2025 attack on Polish energy sites, and the attribution dispute around it, belong in ICS attacks and advisories, 2024 to 2026. For a comparison with a China-attributed actor that mixes espionage and crime, see APT41 public record.

4. Tradecraft is described at the level the advisories use, by technique name and ATT&CK id. The study reproduces no commands, payloads or exploit details.

Some sources did not load on 3 November 2026 and are not cited as opened. justice.gov served a bot-verification page to scripted clients, so the Justice Department press releases of 13 July 2018, 4 October 2018, 19 October 2020, 24 March 2022, 9 May 2023 (Operation MEDUSA), 5 September 2024 and 3 October 2024 were not read; the indictments are cited instead through the FBI wanted notices, which are primary government documents that restate each charge, court and date; the unit assignments of Morenets and Andrienko come from their individual FBI wanted posters, read from archived copies on 10 November 2026 [106] [107]; and one Justice Department release (15 February 2024) was read from an archived copy of the same URL. NSA PDFs on media.defense.gov returned 403, including the 2021 GRU brute-force advisory (only the CISA bulletin announcing it was opened) and the 2020 Drovorub advisory. The NCSC's October 2018 GRU statement has been removed. Ukraine's SSU statement on Gamaredon, the German government's May 2024 APT28 attribution, the US State Department's Viasat page, the Dutch AIVD and MIVD Laundry Bear report, HHS's February 2024 brief on Russian actors and the New Jersey court opinion in the Merck case did not load either. Where a claim rests on a secondary account of one of these, it is graded Reported.

Part 1: the services, the units and the names

The unit map as governments draw it

Service	Unit as governments name it	Community names cited by the same documents	ATT&CK object	Sources
GRU	85th Main Special Service Center (GTsSS), military unit 26165	APT28, Fancy Bear, Forest Blizzard, BlueDelta, Sofacy, Pawn Storm, Sednit, Tsar Team	G0007 APT28	[30] [55] [60] [80]
GRU	Main Center for Special Technologies (GTsST), military unit or field post 74455	Sandworm, APT44, Seashell Blizzard, ELECTRUM, Voodoo Bear, Telebots	G0034 Sandworm Team	[30] [10] [15] [71] [82]
GRU	161st Specialist Training Center, military unit 29155	Cadet Blizzard, Ember Bear, UNC2589, UAC-0056, DEV-0586	G1003 Ember Bear	[51] [86]
SVR	Foreign Intelligence Service	APT29, Cozy Bear, the Dukes, NOBELIUM, Midnight Blizzard	G0016 APT29	[19] [22] [30] [45] [81]
FSB	Center 16, military unit 71330	Berserk Bear, Energetic Bear, Dragonfly, Crouching Yeti, Ghost Blizzard, Static Tundra; and a unit within Center 16 running the Snake and Turla toolset (Venomous Bear, Secret Blizzard)	G0035 Dragonfly; G0010 Turla	[30] [38] [56] [83] [84]
FSB	Center 18 (Center for Information Security), military unit 64829	Star Blizzard, SEABORGIUM, Callisto Group, COLDRIVER, TA446; Gamaredon (Primitive Bear, Armageddon) per Ukraine's government only	G1033 Star Blizzard; G0047 Gamaredon Group	[30] [40] [85]
Ministry of Defense	Central Scientific Research Institute of Chemistry and Mechanics (TsNIIKhM)	TEMP.Veles, XENOTIME; the Triton (TRISIS, HatMan) operators	G0088 TEMP.Veles	[17] [30]

France named a further GRU unit. Its April 2025 statement links GRU Unit 20728 to pressure on Ukrainian infrastructure, while UK and US documents tie APT28 itself to Unit 26165 (**Confirmed** [53] [55] [60]).

How ATT&CK words each attribution

ATT&CK, the community's shared catalog of adversary behavior, carries its own attribution sentence for each group. Read side by side with the government record, the wording falls into four tiers (**Confirmed** for each sentence [80] to [90]; the tiering is **Inferred**).

Tier	Groups	ATT&CK wording
Named service in ATT&CK's own text	APT28 (GRU 85th GTsSS, unit 26165), Sandworm Team (GRU GTsST, unit 74455), Ember Bear (GRU 161st Specialist Training Center, Unit 29155), APT29 (SVR), Turla (FSB), Dragonfly (FSB Center 16)	Attributed to the named unit or service [80] [82] [86] [81] [83] [84]

Tier	Groups	ATT&CK wording
Service named only as another government's claim	Gamaredon Group	"Suspected Russian"; ATT&CK carries the November 2021 Ukrainian government attribution to FSB Center 18 rather than its own [85]
Russia-nexus, no service named	Star Blizzard (originating in Russia, campaigns aligned to Russian state interests), Saint Bear (Russian-nexus, since early 2021), Winter Vivern (linked to Russian and Belorussian interests), TEMP.Veles (Russia-based), ALLANITE (suspected Russian, ATT&CK for ICS only)	[87] [88] [89] [94] [95]
Russia-based criminal	Wizard Spider (G0102), Indrik Spider (G0119, Evil Corp)	Criminal groups; Indrik Spider changed tactics after 2019 US sanctions and an indictment [93]

Three groups ATT&CK describes only by language (FIN5, whose actors "likely speak Russian"; Nomadic Octopus, "Russian-speaking"; RedCurl, "allegedly a Russian-speaking" actor) [93] are left out: language is not attribution.

Vendor names in common use

Microsoft's "Blizzard" names appear throughout the sources opened, so they are given here with the other names each ATT&CK object lists (**Confirmed** from the library's copy of the ATT&CK alias lists [93] and the vendor posts cited).

ATT&CK object	Blizzard name	Other names on the ATT&CK object
APT28 G0007	Forest Blizzard	Fancy Bear, STRONTIUM, Sofacy, Sednit, Pawn Storm, Tsar Team, IRON TWILIGHT, SNAKEMACKEREL, FROZENLAKE, GruesomeLarch
Sandworm Team G0034	Seashell Blizzard	APT44, ELECTRUM, IRIDIUM, Telebots, Voodoo Bear, IRON VIKING, FROZENBARENTS
Ember Bear G1003	Cadet Blizzard	DEV-0586, UNC2589, Bleeding Bear, Frozenvista, UAC-0056
APT29 G0016	Midnight Blizzard	Cozy Bear, The Dukes, NOBELIUM, YTTRIUM, UNC2452, UNC3524, Dark Halo, IRON RITUAL, IRON HEMLOCK
Turla G0010	Secret Blizzard	Snake, Venomous Bear, Krypton, Waterbug, WhiteBear, IRON HUNTER
Dragonfly G0035	Ghost Blizzard	Energetic Bear, Berserk Bear, Crouching Yeti, BROMINE, TEMP.Isotope, DYMALLOY, IRON LIBERTY
Star Blizzard G1033	Star Blizzard	SEABORGIUM, Callisto Group, COLDRIVER, TA446
Gamaredon Group G0047	Aqua Blizzard	Primitive Bear, ACTINIUM, Armageddon, Shuckworm, IRON TILDEN

Two naming traps are worth knowing. ATT&CK lists UAC-0056 as an alias of both Ember Bear and Saint Bear (G1031), while treating the two as distinct groups, and AA24-249A uses UAC-0056 for Unit 29155 (**Confirmed**

[93] [86] [51]). And AA24-249A warns that vendor names may not map one-to-one to its own attribution (**Confirmed** [51]). The FBI's 2025 notice makes the same point for FSB Center 16, calling Berserk Bear and Dragonfly separate but related clusters (**Confirmed** [56]).

Part 2: the groups one by one

APT28 (GRU 85th GTsSS, Unit 26165)

ATT&CK attributes APT28 to GRU military unit 26165, active since at least 2004; the library data lists 93 Enterprise techniques and 29 software entries, and its only ATT&CK campaign is `C0051`, the Nearest Neighbor campaign of February 2022 to November 2024 (**Confirmed** [80]). The live page lists 31 software ids; the library's copy lacks two, netsh (`S0108`) and the Mobile-domain X-Agent for Android (`S0314`, no library page) (**Confirmed** [80] [93]).

FireEye's first public report, on 27 October 2014, said APT28 was most likely sponsored by the Russian government from Moscow and active since at least 2007 against governments, militaries and security organisations (**Confirmed** [65]). The UK's GRU profile attributes to Unit 26165 the April to May 2015 Bundestag hack, TV5Monde (2015), the 2016 DNC and DCCC intrusions, the 2017 French election, the 2018 intrusions into the OPCW, the Spiez laboratory and DSTL, and Paris 2024 (**Confirmed** [60]). The 2016 election thread is covered in the US record: the December 2016 GRIZZLY STEPPE report confirms APT28 entered a US political party in spring 2016 (**Confirmed** [2]), CrowdStrike placed FANCY BEAR in the DNC network from April 2016 with possible GRU affiliation (**Confirmed** [66]), and the July 2018 indictment placed defendant Lukashev in Unit 26165 (**Confirmed** [8]).

APT28 also attacked sport. WADA confirmed in September 2016 that Tsar Team (APT28, Fancy Bear) reached its ADAMS database through an IOC-created Rio 2016 account, probably by spear-phishing, and published confidential athlete medical data, including Therapeutic Use Exemptions (**Confirmed** [1]). A Western District of Pennsylvania grand jury indicted seven GRU officers on 3 October 2018 for intrusions from 2014 to May 2018 into USADA, WADA and other victims; defendant Morenets is listed in Unit 26165 (**Confirmed** [9] [106]).

From 2021 the record turns to network devices and mail. APT28 exploited CVE-2017-6742 on Cisco routers to deploy Jaguar Tooth, hitting about 250 Ukrainian victims and US government institutions; the NCSC and partners assess "almost certainly" Unit 26165, with high confidence (**Confirmed** [36] [37]). In January 2024 a court-authorized operation neutralized hundreds of SOHO routers that Unit 26165 had repurposed from a network criminals built with Moobot malware (**Confirmed**, archived DOJ release [44]), and an 11-country advisory of 27 February 2024 described APT28's use of compromised Ubiquiti EdgeRouters to harvest credentials, collect NTLMv2 digests, proxy traffic and host spear-phishing pages (**Confirmed** [46]). Czechia attributed attacks on its institutions using a 2023 Outlook zero-day to APT28 on 3 May 2024 (**Confirmed** [47]). France formally condemned the GRU's use of APT28 against about a dozen French entities since 2021 on 29 April 2025, and ANSSI's companion report covers French government, diplomatic and research targets from 2021 to 2024 while naming no service (**Confirmed** [53] [54]).

AA25-141A, published in May 2025 by agencies from eleven countries and corrected in April 2026, reports that Unit 26165 has targeted Western logistics and technology companies since 2022 across 13 countries, including air traffic management, and targeted IP cameras at borders and rail stations from March 2022, 81% of them in Ukraine; the camera activity is assessed as "likely connected" (**Confirmed** [55]). In April 2026 the

FBI reported that since at least 2024 APT28 had compromised SOHO routers, including TP-Link devices through CVE-2023-50224, and hijacked DHCP and DNS to run adversary-in-the-middle attacks on Outlook Web Access before narrowing to military, government and critical-infrastructure users (**Confirmed** [59]).

Sandworm Team (GRU GTsST, Unit 74455)

ATT&CK attributes Sandworm Team to GRU unit 74455, active since at least 2009, with 79 Enterprise, 4 ICS and 3 Mobile techniques, 27 software entries and three campaigns: `C0028` (December 2015 to January 2016), `C0025` (December 2016) and `C0034` (June to October 2022) (**Confirmed** [82]). The live page also shows CaddyWiper (`S0693`) and the Mobile-domain CHEMISTGAMES, which are not in the library's G0034 software list (**Confirmed** [82] [93]).

The operational story lives in SANDWORM_OPERATIONS.md. In summary: the UK's GRU profile attributes to Unit 74455 BlackEnergy (2015, when 230,000 people lost power), Industroyer (2016), NotPetya and BadRabbit (2017), Georgia (2018) and Olympic Destroyer (2018) (**Confirmed** [60]); the White House and the UK both attributed NotPetya to the Russian military on 15 February 2018 (**Confirmed** [3] [4]); a Western District of Pennsylvania grand jury indicted six GRU officers, among them Andrienko of Unit 74455, on 15 October 2020 for Ukraine, the French campaign, Georgia, NotPetya, the 2018 Winter Olympics and nerve-agent investigations (**Confirmed** [14] [107]); the US and UK attributed the 28 October 2019 Georgia attack to the GTsST on 20 February 2020 (**Confirmed** [10] [11]); and the UK exposed GTsST reconnaissance of Tokyo 2020 and confirmed the Pyeongchang Olympic Destroyer false flag on 19 October 2020 (**Confirmed** [15]). Mandiant graduated the group to APT44 in April 2024, "sponsored by Russian military intelligence" (**Confirmed** [71]). Ukraine's SBU named Unit 74455 for the December 2023 Kyivstar outage, which affected 24 million customers; the UK profile records no UK attribution of that attack (**Reported** [60]). AA25-343A adds that Unit 74455 likely supported the creation of the hacktivist front CARR and funded its DDoS tools through at least September 2024 (**Confirmed** [57]).

Ember Bear and Cadet Blizzard (GRU Unit 29155)

ATT&CK links Ember Bear to the GRU 161st Specialist Training Center (Unit 29155), active since at least 2020, and credits it with the WhisperGate (`S0689`) wiper attacks on Ukraine in early 2022; it has 47 techniques and 11 software entries (**Confirmed** [86]). AA24-249A, from the FBI, CISA, NSA and partners on 5 September 2024, attributes operations since 2020 to Unit 29155, including WhisperGate from 13 January 2022, and says the unit is separate from Units 26165 and 74455 (**Confirmed** [51]). The FBI counted more than 14,000 domain-scanning instances across at least 26 NATO members, and the listed target sectors include government services, financial services, transportation, energy and healthcare in NATO and EU members and in Central American and Asian countries (**Confirmed** [51]). The advisory notes that WhisperGate is not unique to Unit 29155 and that a related sample was seen at a US-based victim (**Confirmed** [51]). Five Unit 29155 officers and a civilian, Amin Stigal, are wanted for activity from December 2020 to August 2024, and Rewards for Justice offers up to $10 million (**Confirmed** [52]). The UK profile attributes Estonia (2020) and WhisperGate (2022) to the unit, and the UK said in May 2022 that Russian military intelligence was "almost certainly" involved in WhisperGate (**Confirmed** [60] [31]).

APT29 (SVR)

ATT&CK attributes APT29 to the SVR, operating since at least 2008 against European and NATO government networks, research institutes and think tanks, with 66 techniques, 49 software entries and two campaigns: C0023 Operation Ghost (September 2013 to October 2019) and C0024, the SolarWinds Compromise (August 2019 to January 2021) (**Confirmed** [81]). The GRIZZLY STEPPE report confirms APT29 entered a US political party in summer 2015, and CrowdStrike found COZY BEAR in the DNC from summer 2015 with no sign of collaboration with FANCY BEAR (**Confirmed** [2] [66]).

The SolarWinds story belongs in SOLARWINDS_SUNBURST.md. In summary: on 15 April 2021 the White House formally named the SVR, citing the Intelligence Community's high confidence and more than 16,000 affected systems; Treasury sanctioned six technology companies that support the intelligence services' cyber programs; and the UK assessed it highly likely that the SVR compromised SolarWinds Orion (**Confirmed** [19] [20] [21]).

The SVR's healthcare record starts in 2020. On 16 July 2020 the NCSC and Canada's CSE assessed APT29 as almost certainly part of the Russian intelligence services and said it had targeted COVID-19 vaccine development in Canada, the US and the UK with WellMess (S0514) and WellMail (S0515); the NSA agreed and CISA endorsed the technical detail (**Confirmed** [12] [13]). AA21-116A then equated SVR cyber actors with APT29 and reported WELLMESS intrusions that went after vaccine research repositories and Active Directory servers (**Confirmed** [22]). In December 2023, the FBI, CISA, NSA, Poland's SKW, CERT Polska and the NCSC reported that the SVR had exploited JetBrains TeamCity CVE-2023-42793 at scale since September 2023, compromising a few dozen companies and more than 100 devices; victims included software firms making billing and medical-device software, and the agencies judged the exploitation opportunistic (**Confirmed** [42]).

Microsoft's breach disclosure follows. Midnight Blizzard password-sprayed a legacy test tenant account from late November 2023 and read senior leadership email; Microsoft detected it on 12 January 2024, and its posts say "Russian state-sponsored" without naming the SVR (**Confirmed** [68]). The 8 March 2024 update reports access to some source code repositories and internal systems and password sprays up to ten times higher in February 2024 (**Confirmed** [70]). HPE's 8-K says a suspected nation-state actor believed to be Midnight Blizzard took data from its cloud email from May 2023, and that HPE was notified on 12 December 2023 (**Confirmed** [43]). AA24-057A, from the NCSC and seven partner agencies on 26 February 2024, assesses that APT29 is almost certainly part of the SVR and is shifting to initial access through cloud services (**Confirmed** [45]). Microsoft saw Midnight Blizzard's RDP-file spear-phishing first on 22 October 2024, against thousands of targets in more than 100 organisations, and noted the US and UK attribution to the SVR (**Confirmed** [73]). On 31 July 2026 Microsoft reassessed Storm-2372, a device-code phishing actor active since August 2024 against sectors including health, as an initial-access sub-cluster of Midnight Blizzard; in February 2025 it had called the actor Russia-aligned at moderate confidence (**Confirmed** [75]).

Turla and Snake (FSB Center 16)

ATT&CK attributes Turla to the FSB, with victims in more than 50 countries since at least 2004, including government, embassies, military, education, research and pharmaceutical companies; it has 68 techniques, 30 software entries and no campaign objects (**Confirmed** [83]). Kaspersky's Epic Turla report of August 2014 found several hundred infected computers in more than 45 countries in the same sectors, with earliest dated artifacts from September 2012, and named no sponsor (**Confirmed** [64]).

AA23-129A, from nine agencies on 9 May 2023, says the FSB began developing Snake as "Uroburos" in late 2003, with first versions around early 2004, and attributes Snake to a known unit within Center 16 that also runs Turla tools; it calls Snake the most sophisticated espionage tool in the FSB's arsenal and rates memory analysis as the most effective detection (**Confirmed** [38]). Microsoft later reported Secret Blizzard taking over the infrastructure of the Pakistan-based Storm-0156 from November 2022 to reach Afghan government and Indian military targets (**Confirmed** [74]), and using an ISP-level adversary-in-the-middle position to push ApolloShadow at foreign embassies in Moscow, seen in February 2025 (**Confirmed** [77]). Both posts rely on CISA's FSB Center 16 attribution. The library's Turla software pages include Uroburos (`S0022`), Carbon (`S0335`), Kazuar (`S0265`), ComRAT (`S0126`), LightNeuron (`S0395`) and TinyTurla (`S0668`) (**Confirmed** [93]).

Dragonfly, Berserk Bear and Static Tundra (FSB Center 16)

ATT&CK attributes Dragonfly to FSB Center 16, active since at least 2010 against defense, aviation, government, ICS-related and critical infrastructure targets, with 56 Enterprise and 2 ICS techniques and 10 software entries (**Confirmed** [84]). Its government record is a decade long. TA18-074A, of 15 March 2018, described a multi-stage campaign since at least March 2016 by "Russian government cyber actors" against US government entities and the energy, nuclear, commercial facilities, water, aviation and critical manufacturing sectors, staging through small commercial facilities and collecting ICS information; it names Dragonfly only through Symantec's reporting (**Confirmed** [5]). AA20-296A, of 22 October 2020, reported Berserk Bear targeting dozens of US state, local, territorial and tribal government and aviation networks since at least September 2020, with data taken from at least two servers (**Confirmed** [16]). On 24 March 2022 the UK and US exposed Center 16 as the group behind a global campaign against critical infrastructure, noting the FBI indictment for targeting the Wolf Creek nuclear plant in Kansas in 2017 (**Confirmed** [28]).

The network-device thread runs from 2018 to 2026. The FBI's August 2025 notice attributes to Center 16 (Talos's Static Tundra) exploitation of Cisco Smart Install CVE-2018-0171 and of SNMP on end-of-life devices, with configuration files collected from thousands of devices at US critical-infrastructure entities and an interest in ICS protocols; it says the unit has compromised network devices for over a decade (**Confirmed** [56]). AA26-194A, from 19 agencies on 13 July 2026, describes Center 16 opportunistically abusing SNMP default community strings and Cisco flaws CVE-2018-0171 and CVE-2008-4128 to copy router configurations worldwide, and names Healthcare and Public Health among the sectors most at risk (**Confirmed** [62]). On the same day the UK and EU member states formally attributed the failed December 2025 attack on Poland's energy sector to FSB Centre 16, saying it could have cut power to 500,000 citizens (**Confirmed** [61] [63]).

Star Blizzard (FSB Center 18)

ATT&CK describes Star Blizzard as originating in Russia with campaigns aligned to Russian state interests; the description does not name the FSB, which appears only in the title of the joint advisory it cites (**Confirmed** [87]). That advisory, AA23-341A of 7 December 2023, from the NCSC and seven partners, assesses Star Blizzard as almost certainly subordinate to FSB Centre 18 and describes its targeting of academia, defense, government, NGOs, think tanks and politicians, mostly in the UK and US, expanding in 2022 to defense-industrial targets and DOE facilities (**Confirmed** [40]). The UK statement of the same day attributes the targeting of UK parliamentarians from 2015 to 2023, the 2018 Institute for Statecraft hack, the 2019 UK-US trade document leak and the compromise of the Institute founder's account from December 2021 (**Confirmed** [41]). A Northern District of California grand jury indicted FSB officer Ruslan Peretyatko and Callisto Group affiliate Andrey Korinets for a spear-phishing campaign seeking US intelligence on defense, foreign affairs and nuclear research (**Confirmed** [39]). Microsoft seized 66 Star Blizzard domains by court order and the Justice Department seized 41 more in October 2024; Microsoft counted 82 customers targeted from January 2023 (**Confirmed** [72]). Google's threat intelligence group attributes LOSTKEYS and its 2025 replacements NOROBOT, YESROBOT and MAYBEROBOT, delivered through fake CAPTCHA lures, to COLDRIVER, which it calls Russian state-sponsored without naming an agency (**Confirmed** [78]). The only ATT&CK software object is Spica (`S1140`) (**Confirmed** [87]).

Gamaredon Group (FSB Center 18, per Ukraine)

ATT&CK describes Gamaredon as a suspected Russian group targeting Ukrainian military, law enforcement, judiciary and NGOs since 2013, and carries the November 2021 Ukrainian government attribution to FSB Center 18 rather than its own; it has 70 techniques and 6 software entries (**Confirmed** [85]). The five-government AA22-110A describes PRIMITIVE BEAR only as industry reporting and does not attribute it (**Confirmed** [30]). The SSU primary did not load, so the Center 18 link is **Reported** here [99].

TEMP.Veles and Triton (TsNIIKhM)

ATT&CK describes TEMP.Veles as Russia-based and tied to TRITON, with 0 Enterprise and 2 ICS techniques and two campaigns: `C0032`, the IT foothold of October 2014 to January 2017, and `C0030`, the Triton safety-instrumented-system attack of June to August 2017 (**Reported**, from the library's copy of ATT&CK [94]). Treasury designated TsNIIKhM on 23 October 2020 for its connection to Triton and its role in the August 2017 attack on a Middle East petrochemical facility, citing reported 2019 probing of at least 20 US electric utilities (**Confirmed** [17]). The FBI notice for Evgeny Gladkikh of TsNIIKhM's Applied Development Center lists a charge of conspiracy to damage an energy facility from August 2014 to July 2018, with safety instrumented systems as the target, in the case announced on 24 March 2022 (**Confirmed** [29]). The library has no generated page for the Triton software object `S1009`, an ICS-domain entry (**Confirmed** [100]).

Russia-nexus groups with no named service in the sources opened

These groups are tied to Russia by ATT&CK or a vendor, but no government source opened for this study names a service for them.

Group	What the record says	Grade and sources
Saint Bear `G1031`	Russian-nexus actor active since early 2021 against Ukraine and Georgia, using Saint Bot (`S1018`) and OutSteel (`S1017`); 18 techniques; ATT&CK keeps it distinct from Ember Bear	Confirmed [88] [86]

Group	What the record says	Grade and sources
Winter Vivern G1035	No explicit state attribution; linked to Russian and Belorussian interests; active since at least 2020 against European government and NGO entities; 27 techniques, no software	Confirmed [89]
Void Blizzard (Laundry Bear)	Microsoft assesses with high confidence that it is Russia-affiliated, active since at least April 2024, targeting sectors including healthcare, with a Ukrainian aviation target in October 2024 and NGO adversary-in-the-middle phishing in April 2025; Dutch AIVD and MIVD and the FBI collaborated; no ATT&CK group id found	Confirmed [76]
Laundry Bear and the Dutch police	Dutch services tie Laundry Bear to the September 2024 Dutch police breach and call it extremely likely Russian state supported	Reported (press account [103]; the AIVD and MIVD primary did not load)
ALLANITE (G1000)	ATT&CK for ICS only: a suspected Russian group targeting US and UK electric utilities, 4 ICS techniques; the library has no generated page for it	Reported [95]

Russia-based criminal and hacktivist groups

These are not state units, but the government record ties several of them to Russia and to healthcare, so defenders meet them in the same threat models.

- **Wizard Spider** (G0102): TrickBot (S0266), Conti (S0575) and Ryuk (S0446); ATT&CK lists targets ranging to hospitals, with 64 techniques and 22 software entries (**Reported** [93]). Treasury calls Trickbot a Russia-based gang whose current members are associated with the Russian Intelligence Services (**Confirmed** [34]).
- **Indrik Spider** (G0119, Evil Corp): Dridex (S0384), BitPaymer (S0570) and WastedLocker (S0612); ATT&CK records that it changed tactics after the 2019 US sanctions and indictment (**Reported** [93]).
- **LockBit**: Treasury designated Russian national Dmitry Khoroshev as leader of the Russia-based LockBit group on 7 May 2024 (**Confirmed** [48]).
- **KillNet**: HC3 says KillNet has been active since January 2022 and that its ties to the FSB or SVR are unconfirmed (**Confirmed** [33]).
- **CARR and NoName057(16)**: AA25-343A assesses that GRU Unit 74455 likely supported CARR's creation and funded its DDoS tools, and that the Kremlin-linked CISM created NoName057(16); the named targets were water, food and agriculture, and energy (**Confirmed** [57]).

Part 3: the operational record

Timeline

One row per dated operation, report or government action. "Attributed to" uses each source's own name and hedge. Sandworm and SolarWinds rows are kept short; their home studies carry the detail.

Date	Attributed to	Event, report or action	Targets and sectors	Grade	Sources
Late 2003 to early 2004	FSB (Center 16 unit)	Snake development begins as "Uroburos"; first versions around early 2004	Government, research, journalists in 50+ countries (later)	Confirmed	[38]
From 2004	APT28; Turla	Earliest activity ATT&CK records for each group	Government, embassies, military, research, pharmaceutical (Turla)	Confirmed	[80] [83]
From 2008	APT29 (SVR)	Earliest activity ATT&CK records	European and NATO governments, research institutes, think tanks	Confirmed	[81]
From 2009, 2010, 2013	Sandworm; Dragonfly; Gamaredon	Earliest activity ATT&CK records for each	Ukraine; critical infrastructure; Ukrainian military, law enforcement, judiciary and NGOs	Confirmed	[82] [84] [85]
2012-09 to 2014-08	Turla (Kaspersky: no sponsor named)	Epic Turla operation	Several hundred computers, 45+ countries; government, embassies, military, education, research, pharmaceutical	Confirmed	[64]
2013-09 to 2019-10	APT29	Campaign C0023 Operation Ghost	Government targets	Confirmed	[81] [94]
2014 to 2018-05	GRU Unit 26165	Intrusions into USADA, WADA and other anti-doping and sports bodies (indicted 2018-10-03)	Sport, anti-doping	Confirmed	[9] [106]
2014-08 to 2018-07	TsNIIKhM employee	Conspiracy to damage an energy facility, targeting safety instrumented systems (charged, announced 2022-03-24)	Energy	Confirmed	[29]
2014-10 to 2017-01	TEMP.Veles	Campaign C0032 (the Triton IT foothold)	Petrochemical	Reported	[94]
2014-10-27	APT28 (FireEye: most likely Russian government)	First public APT28 report	Governments, militaries, security organisations	Confirmed	[65]
2015-04 to 2015-05	GRU Unit 26165	German Bundestag intrusion	Parliament	Confirmed	[60]
2015	GRU Unit 26165 (UK); APT28 (France)	TV5Monde attack	Broadcaster	Confirmed	[60] [53]
2015 summer	APT29	Intrusion into a US political party (the DNC)	Political	Confirmed	[2] [66]
2015-12	GRU Unit 74455	BlackEnergy attack on Ukraine's grid, 230,000 people without power (campaign C0028); see the Sandworm study	Electric power	Confirmed	[60] [82]

Date	Attributed to	Event, report or action	Targets and sectors	Grade	Sources
2016-03 onward	Russian government cyber actors (Dragonfly per Symantec)	Multi-stage campaign staging through small commercial facilities, collecting ICS information (alert TA18-074A)	US energy, nuclear, commercial facilities, water, aviation, critical manufacturing	Confirmed	[5]
2016 spring (April)	APT28	Intrusion into the DNC; CrowdStrike report 2016-06-14	Political	Confirmed	[2] [66]
2016 (disclosed 2016-09-13)	Tsar Team (APT28)	Access to WADA's ADAMS database through a Rio 2016 account; athlete medical data published	Sport, athlete medical data	Confirmed	[1]
2016	GRU Unit 26165	DNC and DCCC intrusions (UK attribution); 2016 election hack-and-leak (indicted 2018-07-13)	Political, election infrastructure	Confirmed	[60] [7]
2016-12	GRU Unit 74455	Industroyer attack on Kyiv's grid (campaign C0025); see the Sandworm study	Electric power	Confirmed	[60] [82]
2016-12-29	APT28 and APT29 (US government)	GRIZZLY STEPPE joint analysis report	Political	Confirmed	[2]
2017	FSB Center 16	Targeting of the Wolf Creek nuclear plant in Kansas (FBI indictment)	Nuclear	Confirmed	[28]
2017	GRU Unit 26165 (UK); Unit 74455 (2020 indictment)	French presidential election targeting	Political	Confirmed	[60] [14]
2017	FSB employees, one of them in Center 18	Indictment recorded by AA22-110A	Email accounts of US government and military personnel and private organizations	Confirmed	[30]
2017-06	GRU Unit 74455 (Russian military per US and UK)	NotPetya; see the Sandworm study	Worldwide; a US pharmaceutical manufacturer and US hospital record systems per HC3	Confirmed	[3] [4] [32] [60]
2017-06 to 2017-08	TEMP.Veles; TsNIIKhM (Treasury)	Triton attack on a safety instrumented system at a Middle East (UK: Saudi) petrochemical plant (campaign C0030)	Petrochemical	Confirmed	[17] [28] [94]
2017	GRU Unit 74455	BadRabbit	Not stated in the source	Confirmed	[60]
2018 (Pyeongchang Games)	GRU Unit 74455	Olympic Destroyer at Pyeongchang, a false flag posing as North Korean and Chinese hackers, and VPNFilter on devices in the Republic of Korea	Olympics	Confirmed	[15] [60]

Date	Attributed to	Event, report or action	Targets and sectors	Grade	Sources
2018-02-15	Russian military	US and UK attribute NotPetya	Worldwide	Confirmed	[3] [4]
2018-03-15	Russian government cyber actors	Alert TA18-074A	US critical infrastructure	Confirmed	[5]
2018-04-16	Russian state-sponsored actors	Alert TA18-106A on routers, switches, firewalls and NIDS; update 2018-04-19 on DNS redirection through GRE tunnels	Network infrastructure worldwide	Confirmed	[6]
2018	GRU Unit 26165	Intrusions against the OPCW, the Spiez laboratory and DSTL	Chemical-weapons investigation	Confirmed	[60]
2018	Star Blizzard (FSB Centre 18)	Institute for Statecraft hack	Think tank	Confirmed	[41]
2018-07-13	12 GRU officers (Units 26165 and 74455)	District of Columbia indictment for 2016 election interference	Political, election boards and suppliers	Confirmed	[7] [8]
2018-10-03	7 GRU officers	Western District of Pennsylvania indictment (announced 2018-10-04)	Anti-doping, sport	Confirmed	[9]
2018	GRU Unit 74455	Attacks on Georgia	Government	Confirmed	[60]
2019	Star Blizzard	UK-US trade document leak	Government	Confirmed	[41]
2019	Indrik Spider (Evil Corp)	US sanctions and indictment	Financial crime	Reported	[93]
2019	Reported probing cited by Treasury in the TsNIIKhM designation	Probing of at least 20 US electric utilities	Electric power	Confirmed (that Treasury cites it)	[17]
2019-10-28	GRU GTsST (Unit 74455)	Disruptive attack on Georgia: several thousand websites and at least two TV broadcasters; US and UK attribution 2020-02-20	Government, media	Confirmed	[10] [11]
2019-08 to 2021-01	APT29 (SVR)	SolarWinds Compromise (campaign C0024); see the SolarWinds study	Government, technology; 16,000+ systems	Confirmed	[19] [81]
2020	APT29	Targeting of COVID-19 vaccine development; advisory 2020-07-16	Healthcare, pharmaceutical research in Canada, the US and the UK	Confirmed	[12] [13]
2020	GRU Unit 29155	Attack on Estonia	Government	Confirmed	[60]
2020-09 onward	Berserk Bear (FSB Center 16)	Intrusions into US state, local, territorial and tribal government and aviation networks (AA20-296A, 2020-10-22)	Government, aviation	Confirmed	[16]

Date	Attributed to	Event, report or action	Targets and sectors	Grade	Sources
2020-10-15	6 GRU officers (Andrienko listed in Unit 74455)	Western District of Pennsylvania indictment	Ukraine, France, Georgia, NotPetya victims, Olympics	Confirmed	[14] [107]
2020-10-19	GRU GTsST	UK exposes reconnaissance against Tokyo 2020 organisers, logistics services and sponsors	Olympics	Confirmed	[15]
2020-10-23	TsNIIKhM	Treasury designation under CAATSA section 224	Petrochemical, energy	Confirmed	[17]
2020-10-28	Cybercrime (no attribution)	AA20-302A: increased and imminent threat to US hospitals from TrickBot, BazarLoader, Ryuk and Conti	Healthcare	Confirmed	[18]
2020-11	Strontium among three nation-state actors (Microsoft)	Attacks on seven companies researching COVID-19 vaccines and treatments	Pharmaceutical research in Canada, France, India, South Korea, the US	Confirmed	[67]
2021-03-16 to 2021-05-14	Conti operators (nationality not stated)	Health Service Executive, Ireland: phishing to Conti detonation; all IT switched off	Healthcare	Confirmed	[25]
2021-04-15	SVR	US formal attribution of SolarWinds; Treasury sanctions on six technology companies; UK attribution	Government, technology	Confirmed	[19] [20] [21]
2021-04-26	SVR (APT29)	AA21-116A on SVR trends and WELLMESS	Government, think tanks, IT, vaccine research	Confirmed	[22]
2021-05-20	Conti	FBI FLASH: at least 16 attacks on US healthcare and first-responder networks	Healthcare, first responders	Confirmed	[23]
2021	APT28 (GRU Unit 26165)	Jaguar Tooth on Cisco routers through CVE-2017-6742; advisory 2023-04-18	About 250 Ukrainian victims, US government	Confirmed	[36] [37]
2021-07-01	GRU 85th GTsSS	Joint advisory on a global brute-force campaign (bulletin opened; the advisory PDF did not load)	Not checked	Confirmed	[24]
2021-11	Gamaredon (FSB Center 18, per Ukraine)	Ukrainian government attribution	Ukrainian military, law enforcement, judiciary	Reported	[85] [99]
2021 to 2024	APT28 (GRU, per France)	Attacks on about a dozen French entities, including a Paris 2024 sports body	Government, diplomatic, research, sport	Confirmed	[53] [54]
2021-12 onward	Star Blizzard	Compromise of the Institute for Statecraft founder's account	Think tank	Confirmed	[41]

Date	Attributed to	Event, report or action	Targets and sectors	Grade	Sources
2022-01-11	Russian state-sponsored actors	AA22-011A: 13 initial-access CVEs, TTPs and mitigations	US critical infrastructure	Confirmed	[26]
2022-01-13	GRU Unit 29155	WhisperGate against Ukrainian organisations	Government	Confirmed	[51] [31]
2022-02 to 2024-11	APT28	Campaign C0051 Nearest Neighbor: nearby Wi-Fi networks and zero-day CVE-2022-38028	Organisations with expertise on Ukraine	Confirmed	[97]
2022-02-15 to 2022-02-16	GRU (UK)	DDoS on Ukraine's banking sector; UK judgment 2022-02-18	Finance	Confirmed	[27]
2022-02-24	Russia (UK: almost certainly; no unit named)	Viasat KA-SAT disruption about an hour before the invasion, also hitting central European wind farms	Satellite communications, energy	Confirmed	[31]
2022-03	Killnet	DDoS on a US airport	Aviation	Confirmed	[30]
2022-03 onward	GRU Unit 26165	IP-camera targeting at borders and rail stations, 81% in Ukraine	Transport	Confirmed	[55]
2022-03-24	FSB Center 16; TsNIIKhM	UK and US expose Center 16; US charges four Russian government employees for 2012 to 2018 energy-sector intrusions; UK sanctions TsNIIKhM	Energy, nuclear	Confirmed	[28] [29]
2022-04-20	GRU, SVR, FSB, TsNIIKhM	Five Eyes advisory AA22-110A maps services to groups	Critical infrastructure	Confirmed	[30]
2022-05	KillNet	Reported threat against ventilators in British hospitals and the UK health ministry	Healthcare	Confirmed	[33]
2022 onward	GRU Unit 26165	Targeting of Western logistics and technology companies in 13 countries (AA25-141A)	Defense, transport, maritime, air traffic management, IT	Confirmed	[55]
2022-06 to 2022-10	Sandworm	Campaign C0034, 2022 Ukraine electric power attack; see the Sandworm study	Electric power	Confirmed	[82]
2022-11 onward	Secret Blizzard (Turla)	Takeover of Storm-0156 infrastructure (Microsoft post 2024-12-04)	Afghan government, Indian military	Confirmed	[74]
2022-12-08 and 2022-12-16	KillMilk; KillNet	Threat to sell Americans' health data; targeting of Humana Military	Healthcare	Confirmed	[35]

Date	Attributed to	Event, report or action	Targets and sectors	Grade	Sources
2023	APT28 (GRU)	Attacks on Czech institutions through a 2023 Outlook zero-day; Czech attribution 2024-05-03	Government	Confirmed	[47]
2023-01 to 2024-08	Star Blizzard	82 Microsoft customers targeted from January 2023; 30+ civil-society organisations	Civil society, government	Confirmed	[72]
2023-01-28	KillNet and affiliates	Over 90 coordinated DDoS attacks; 55% on health systems with a hospital or Level I trauma centres, 48 US states alleged	Healthcare	Confirmed	[35]
2023-02-09	Trickbot	Treasury sanctions; members associated with Russian Intelligence Services; ambulances diverted from three Minnesota facilities in 2020	Healthcare	Confirmed	[34]
2023-04-18	APT28	NCSC, NSA, CISA and FBI advisory on Jaguar Tooth	Network devices	Confirmed	[36] [37]
2023-05 to 2023-12-12	Midnight Blizzard (HPE: believed to be)	Data taken from HPE's cloud email; 8-K filed 2024-01-19	Technology	Confirmed	[43]
2023-05-09	FSB Center 16	AA23-129A on Snake	Government, research, media in 50+ countries	Confirmed	[38]
2023-09 onward	SVR (APT29)	Exploitation of TeamCity CVE-2023-42793 (AA23-347A, 2023-12-12)	Software firms incl. billing and medical-device software; an energy trade association	Confirmed	[42]
2023-11 to 2024-01-12	Midnight Blizzard	Password spray of a legacy test tenant; senior leadership email read	Technology	Confirmed	[68] [69]
2023-12	GRU Unit 74455 (per Ukraine's SBU)	Kyivstar outage, 24 million customers	Telecommunications	Reported	[60]
2023-12-05	FSB officer Peretyatko; Callisto affiliate Korinets	Northern District of California indictment	Defense, foreign affairs, nuclear research	Confirmed	[39]
2023-12-07	Star Blizzard (FSB Centre 18)	AA23-341A; UK attribution and sanctions	Academia, defense, government, NGOs, think tanks, politicians	Confirmed	[40] [41]
2024-01 (announced 2024-02-15)	GRU Unit 26165	Court-authorized disruption of a SOHO router network built on Moobot	Network devices	Confirmed	[44]
2024-02	Midnight Blizzard	Access to source code repositories; sprays up to tenfold (Microsoft 2024-03-08)	Technology	Confirmed	[70]

Date	Attributed to	Event, report or action	Targets and sectors	Grade	Sources
2024-02-26	SVR (APT29)	AA24-057A on cloud initial access	Aviation, education, law enforcement, local councils, military	Confirmed	[45]
2024-02-27	GRU 85th GTsSS (APT28)	11-country advisory on compromised Ubiquiti EdgeRouters	Network devices	Confirmed	[46]
2024-04 onward	Void Blizzard (Microsoft)	Espionage against critical sectors, including healthcare	Government, defense, healthcare, NGOs	Confirmed	[76]
2024-04-17	Sandworm (Mandiant APT44)	Mandiant graduates Sandworm to APT44	Ukraine, worldwide	Confirmed	[71]
2024-05-07	LockBit leader	Treasury designation of Dmitry Khoroshev	Healthcare among other sectors	Confirmed	[48]
2024-05-10	Black Basta (origin not stated)	AA24-131A: 12 of 16 critical infrastructure sectors incl. healthcare, 500+ organisations	Healthcare and others	Confirmed	[49]
2024-06-06	Midnight Blizzard	HC3 threat actor profile for the health sector	Healthcare	Confirmed	[50]
2024-08 onward	Storm-2372 (from 2026-07-31: a Midnight Blizzard sub-cluster)	Device-code phishing	Sectors including health	Confirmed	[75]
2024-09	Laundry Bear (Dutch services, per press)	Breach of the Dutch police	Law enforcement	Reported	[103]
2024-09-05	GRU Unit 29155	AA24-249A; five officers and a civilian wanted; $10 million reward	Government, finance, transport, energy, healthcare	Confirmed	[51] [52]
2024	GRU Unit 26165	Paris 2024 targeting	Sport	Confirmed	[60]
2024-10-03	Star Blizzard	Microsoft seizes 66 domains and DOJ 41	Civil society	Confirmed	[72]
2024-10-22	Midnight Blizzard	RDP-file spear-phishing, thousands of targets in 100+ organisations	Government, academia, defense, NGOs	Confirmed	[73]
2024 to 2025-02	Secret Blizzard	ISP-level adversary-in-the-middle pushes ApolloShadow at embassies in Moscow	Diplomatic	Confirmed	[77]
2024 to 2026-04-07	GRU 85th GTsSS (APT28)	SOHO router compromise incl. TP-Link CVE-2023-50224; DHCP and DNS hijack for adversary-in-the-middle on Outlook Web Access	Military, government, critical infrastructure	Confirmed	[59]

Date	Attributed to	Event, report or action	Targets and sectors	Grade	Sources
2025-03 to 2025-12	Russian state-sponsored (ATT&CK); no group associated	Campaign C0063, 2025 Poland Wiper Attacks with DynoWiper and LazyWiper	Polish wind, solar, combined heat and power, a manufacturer	Confirmed	[90] [98]
2025-04-29	GRU (France); APT28 (ANSSI)	French formal attribution and ANSSI report CERTFR-2025-CTI-007	Government, diplomatic, research	Confirmed	[53] [54]
2025-05	GRU Unit 26165	AA25-141A, 11 countries (corrected 2026-04-17)	Logistics, technology	Confirmed	[55]
2025-05 to 2025-09	COLDRIVER (GTIG: Russian state-sponsored)	LOSTKEYS disclosed, then NOROBOT, YESROBOT and MAYBEROBOT through fake CAPTCHA lures (post 2025-10-20)	Not checked	Confirmed	[78]
2025-05-27	Void Blizzard	Microsoft disclosure	Healthcare among critical sectors	Confirmed	[76]
2025-08-20	FSB Center 16 (Static Tundra)	FBI PSA on Smart Install CVE-2018-0171 and SNMP exploitation	US critical infrastructure networking devices	Confirmed	[56]
2025-12-09	GRU Unit 74455 (CARR support); CISM (NoName057(16))	AA25-343A on pro-Russia hacktivists	Water, food and agriculture, energy	Confirmed	[57]
2025-12-29	Sandworm (ESET, medium confidence); Static Tundra and Dragonfly overlap (CERT Polska)	RTUs damaged at 30+ wind and PV farms; wipers tried at a CHP plant (blocked by EDR) and a manufacturer; full record in ICS attacks and advisories, 2024 to 2026	Energy, manufacturing	Confirmed	[79] [58]
2026-04-07	GRU 85th GTsSS (APT28)	FBI PSA on router exploitation	Network devices	Confirmed	[59]
2026-07-13	FSB Centre 16	UK and EU formal attribution of the Poland attack; first joint UK-EU cyber sanctions package (24 designations incl. GRU leaders and OOO IMPULS); 19-agency AA26-194A; NCSC call to critical sectors	Energy; Healthcare and Public Health among most at risk	Confirmed	[61] [62] [63]
2026-07-31	Midnight Blizzard	Microsoft reassesses Storm-2372 as its initial-access sub-cluster	Health among sectors	Confirmed	[75]

Reading the record by era

- **2003 to 2014: long-running espionage, quietly named.** The FSB's Snake work and APT28's activity date from 2003 and 2004, but two landmark vendor reports opened for this study (Kaspersky's Epic Turla and FireEye's APT28) date from 2014, and neither named a service with certainty (**Confirmed** [38] [64] [65]).

- **2015 to 2018: disruption and influence.** The Ukrainian grid attacks, NotPetya, the Bundestag, TV5Monde, the DNC, WADA and the OPCW all fall in four years, and the US and UK answered with formal attribution in 2018 and the first GRU indictments (**Confirmed** [60] [3] [4] [7] [9]).
- **2019 to 2021: supply chains, vaccines and naming the SVR.** SolarWinds and the vaccine-research campaign moved the SVR to the front, and the US formally named it in April 2021 (**Confirmed** [19] [12]).
- **2022 to 2024: war-time operations and identity.** WhisperGate, Viasat and the 2022 grid attack run alongside SVR cloud-identity intrusions at Microsoft and HPE and Unit 26165's logistics campaign (**Confirmed** [51] [31] [68] [43] [55]).
- **2025 to 2026: routers, disagreements and joint sanctions.** Network-device campaigns by both the GRU and the FSB, disputed attribution of the Poland attack, and the first joint UK-EU cyber sanctions package close the record (**Confirmed** [56] [59] [79] [58] [61]).

Part 4: the government record

Indictments

Date	Court	Defendants and unit	Conduct charged	Sources
2017 (recorded in AA22-110A)	Not stated in the source	FSB employees, including one Center 18 employee	Accessing email accounts of US government and military personnel and private organizations	[30]
2018-07-13	District of Columbia	12 GRU officers; Lukashev in Unit 26165, Kovalev in Unit 74455	Eleven charged with hacking, stealing documents and staging their release in the 2016 US elections; Osadchuk and Kovalev charged separately for intrusions into state election boards, secretaries of state and election-technology suppliers	[7] [8]
2018-10-03 (announced 2018-10-04)	Western District of Pennsylvania	7 GRU officers; Morenets in Unit 26165	Intrusions from 2014 to May 2018 into USADA, WADA and other victims	[9] [106]
2020-10-15	Western District of Pennsylvania	6 GRU officers; Andrienko in Unit 74455	Ukrainian critical infrastructure, a French political campaign, Georgia, NotPetya, the 2018 Winter Olympics, nerve-agent investigations	[14] [107]
Announced 2022-03-24	Not stated in the source	4 Russian government employees, incl. Gladkikh of TsNIIKhM	Energy-sector intrusions from 2012 to 2018; conspiracy to damage an energy facility (safety instrumented systems)	[29]
2023-12-05	Northern District of California	FSB officer Peretyatko; Korinets (Callisto Group affiliate)	Spear-phishing for US intelligence on defense, foreign affairs and nuclear research	[39]
Announced 2024-09-05	Not stated in the source	5 GRU Unit 29155 officers and civilian Amin Stigal	Activity from December 2020 to August 2024, incl. WhisperGate; up to $10 million reward	[52]

ATT&CK's own summaries of the 2018 and 2020 indictments are consistent with this record; for the October 2018 case ATT&CK counts five Unit 26165 officers and names a US nuclear facility among the victims, where

the FBI notice counts seven GRU officers (**Reported**, as ATT&CK summaries [93]).

Sanctions and disruptions

Date	Actor	Action	Sources
2019	Indrik Spider (Evil Corp)	US sanctions and indictment, after which the group changed tactics	[93]
2020-10-23	TsNIIKhM	US Treasury designation under CAATSA section 224 for its connection to Triton	[17]
2021-04-15	ERA Technopolis, Pasit, SVA, Neobit, AST, Positive Technologies	US Treasury sanctions on companies supporting the GRU, SVR and FSB cyber programs	[20]
2022-03-24	TsNIIKhM	UK sanctions over the 2017 Saudi petrochemical safety-system incident	[28]
2023-02-09	Trickbot members	US Treasury sanctions	[34]
2023-12-07	Peretyatko and Korinets	UK sanctions	[41]
2024-01 (announced 2024-02-15)	GRU Unit 26165 router network	Court-authorized disruption	[44]
2024-05-07	Dmitry Khoroshev (LockBit)	US Treasury designation	[48]
2024-10-03	Star Blizzard	Microsoft court-ordered seizure of 66 domains; DOJ seizure of 41	[72]
2026-07-13	24 individuals and entities incl. GRU leaders and OOO IMPULS	UK designations announced with the first joint UK-EU cyber sanctions package	[61]

Formal attributions outside the indictments

Date	Government	Actor named	Event	Confidence wording	Sources
2016-12-29	United States	APT28 and APT29 (RIS)	Intrusions into a US political party	"confirms"	[2]
2018-02-15	United States; United Kingdom	Russian military	NotPetya	US: launched by the Russian military; NCSC: almost certainly	[3] [4]
2020-02-20	United States; United Kingdom	GRU GTsST (Unit 74455)	Georgia, 28 October 2019	UK: "with the highest level of probability"	[10] [11]
2020-07-16	UK, Canada; US agrees	APT29	COVID-19 vaccine targeting	Almost certainly part of the intelligence services; highly likely sought vaccine IP	[12] [13]
2020-10-19	United Kingdom	GRU GTsST	Tokyo 2020 reconnaissance; Pyeongchang 2018	"reveals" and "confirms"	[15]
2021-04-15	United States; United Kingdom	SVR	SolarWinds Orion	US: high confidence; UK: highly likely	[19] [21]
2022-02-18	United Kingdom	GRU	DDoS on Ukrainian banks	"judges"	[27]
2022-03-24	United Kingdom with the US	FSB Center 16	Global critical-infrastructure campaign	"exposes"	[28]
2022-04-20	Five Eyes	FSB Center 16 and 18, SVR, GRU units 26165 and 74455, TsNIIKhM	Unit map	Center 16: almost certainly (US and UK)	[30]
2022-05-10	United Kingdom	Russia; Russian military intelligence	Viasat; WhisperGate	Almost certainly	[31]
2023-04-18	UK, US	GRU Unit 26165	Jaguar Tooth	Almost certainly, high confidence	[37]
2023-05-09	Nine agencies	FSB Center 16 unit	Snake	Attributes	[38]
2023-12-07	UK and seven partners	FSB Centre 18	Star Blizzard	Almost certainly subordinate; the UK page also says highly likely	[40] [41]
2024-05-03	Czechia	APT28, associated with the GRU	Outlook zero-day attacks on Czech institutions	"Russian state-controlled actor"	[47]
2024-09-05	US and partners	GRU Unit 29155	WhisperGate and global scanning	Attributes	[51]
2025-04-29	France	GRU (APT28)	About a dozen French entities since 2021	Formal condemnation	[53]
2026-07-13	UK and EU member states	FSB Centre 16	Poland energy attack, December 2025	Formal attribution; no confidence level on the NCSC page	[61] [63]

How confidence moved

The same actors reached readers at very different strength over twelve years. APT28 went from FireEye's "most likely" Russian government (2014) to "almost certainly" Unit 26165 (2023). Dragonfly went from "Russian government cyber actors", with the group name only via Symantec (2018), to "almost certainly" FSB Center 16 (2022) to formal UK-EU attribution (2026). Storm-2372 went from moderate-confidence Russia-aligned (2025) to a Midnight Blizzard sub-cluster (2026). Microsoft's own breach posts on Midnight Blizzard said "Russian state-sponsored" and left the SVR naming to governments (**Inferred** from [65] [37] [5] [30] [61] [75] [68]). The UK GRU profile gives the UK's attribution of BlackEnergy and Industroyer as "in 2015" and "in 2016", the event years, while AA22-110A says only that they were previously attributed; the date of the first UK statement was not checked, so the profile may be using event years as shorthand (**Inferred** [60] [30]).

Part 5: exploited vulnerabilities

AA22-011A lists 13 CVEs that Russian state actors used for initial access; AA20-296A, AA23-347A, AA24-249A, AA25-141A and the FBI's August 2025 notice add 13 more, for 26 in all (**Confirmed** [26] [16] [42] [51] [55] [56]). Every one of the 26 is in KEV as of catalog version 2026.11.03, and KEV marks 11 of them for known ransomware campaign use (**Confirmed** [92]). The KEV date and CWE columns below come from the same feed; CVE-2017-6742, CVE-2022-38028, CVE-2023-50224 and CVE-2008-4128, named in other primaries, were checked in the same feed. The NCSC's 2020 vaccine advisory names four of the same CVEs for APT29 (**Confirmed** [12]).

CVE	Product (KEV)	Named by (actor, date)	KEV added	Ransomware use (KEV)	CWE (KEV)
CVE-2018-13379	Fortinet FortiOS	AA22-011A (2022-01-11); AA20-296A Berserk Bear (2020-10-22); NCSC APT29 (2020-07-16)	2021-11-03	Known	CWE-22
CVE-2019-1653	Cisco RV320 and RV325 routers	AA22-011A	2021-11-03	Unknown	CWE-284
CVE-2019-2725	Oracle WebLogic Server	AA22-011A	2022-01-10	Known	CWE-74
CVE-2019-7609	Elastic Kibana	AA22-011A	2022-01-10	Unknown	CWE-94
CVE-2019-9670	Synacor Zimbra Collaboration Suite	AA22-011A; NCSC APT29	2022-01-10	Unknown	CWE-611
CVE-2019-10149	Exim MTA	AA22-011A; AA20-296A	2022-01-10	Unknown	CWE-78
CVE-2019-11510	Ivanti Pulse Connect Secure	AA22-011A; NCSC APT29	2021-11-03	Known	CWE-22
CVE-2019-19781	Citrix ADC and Gateway	AA22-011A; AA20-296A; NCSC APT29; AA21-116A (SVR, described as a zero-day)	2021-11-03	Known	CWE-22
CVE-2020-0688	Microsoft Exchange Server	AA22-011A; AA20-296A	2021-11-03	Known	CWE-287
CVE-2020-4006	VMware (multiple products)	AA22-011A	2021-11-03	Unknown	CWE-78

CVE	Product (KEV)	Named by (actor, date)	KEV added	Ransomware use (KEV)	CWE (KEV)
CVE-2020-5902	F5 BIG-IP	AA22-011A	2021-11-03	Known	CWE-22
CVE-2020-14882	Oracle WebLogic Server	AA22-011A	2021-11-03	Unknown	None listed
CVE-2021-26855	Microsoft Exchange Server	AA22-011A	2021-11-03	Known	CWE-918
CVE-2020-1472	Microsoft Netlogon	AA20-296A Berserk Bear	2021-11-03	Known	CWE-330
CVE-2021-33044	Dahua IP camera firmware	AA24-249A Unit 29155 (2024-09-05)	2024-08-21	Unknown	CWE-287
CVE-2021-33045	Dahua IP camera firmware	AA24-249A	2024-08-21	Unknown	CWE-287
CVE-2022-26134	Atlassian Confluence	AA24-249A	2022-06-02	Known	CWE-917
CVE-2022-26138	Atlassian Confluence	AA24-249A	2022-07-29	Unknown	CWE-798
CVE-2022-3236	Sophos Firewall	AA24-249A	2022-09-23	Unknown	CWE-94
CVE-2023-23397	Microsoft Office (Outlook NTLM)	AA25-141A Unit 26165 (2025-05)	2023-03-14	Unknown	CWE-294
CVE-2023-38831	RARLAB WinRAR	AA25-141A	2023-08-24	Known	CWE-351
CVE-2020-12641	Roundcube Webmail	AA25-141A	2023-06-22	Unknown	CWE-78
CVE-2020-35730	Roundcube Webmail	AA25-141A	2023-06-22	Unknown	CWE-79
CVE-2021-44026	Roundcube Webmail	AA25-141A	2023-06-22	Unknown	CWE-89
CVE-2023-42793	JetBrains TeamCity	AA23-347A SVR (2023-12-12)	2023-10-04	Known	CWE-288
CVE-2018-0171	Cisco IOS and IOS XE (Smart Install)	FBI PSA, FSB Center 16 (2025-08-20); AA26-194A (2026-07-13)	2021-11-03	Unknown	CWE-20
CVE-2017-6742	Cisco IOS and IOS XE	NCSC and partners, APT28 Jaguar Tooth (2023-04-18)	2023-04-19	Unknown	CWE-119
CVE-2022-38028	Microsoft Windows	ATT&CK campaign C0051, APT28 (zero-day)	2024-04-23	Unknown	None listed
CVE-2023-50224	TP-Link TL-WR841N	FBI PSA, APT28 (2026-04-07)	2025-09-03	Unknown	CWE-290
CVE-2008-4128	Cisco IOS	AA26-194A, FSB Center 16 (2026-07-13)	2026-07-13	Unknown	CWE-352

Sources: [26] [16] [12] [22] [51] [55] [42] [56] [62] [37] [97] [59] [92].

AA24-249A adds that Unit 29155 obtained, but was not seen exploiting, scripts for CVE-2020-1472, CVE-2021-26084, CVE-2021-3156, CVE-2021-4034 and CVE-2022-27666 (**Confirmed** [51]).

Three readings for a vulnerability-management program follow (**Inferred** from the table).

1. **KEV is the right floor, and it is not enough on its own.** All 26 advisory CVEs are in KEV, and the earliest dateAdded values are the catalog's launch day, so a KEV-first program would have caught them.

But CVE-2018-0171 sat in KEV from 3 November 2021 and was still being exploited by FSB Center 16 in August 2025 [92] [56]: the gap was patching, not knowledge.

2. **The edge is the target class.** VPNs and gateways (Fortinet, Pulse, Citrix, F5, Sophos), mail and collaboration (Exchange, Zimbra, Roundcube, Outlook, Confluence), CI/CD (TeamCity), cameras (Dahua) and routers (Cisco, TP-Link) make up nearly the whole list.
3. **Age is no comfort.** CVE-2008-4128 entered KEV on the day of AA26-194A, 13 July 2026, eighteen years after its CVE year [92] [62]. Long-lived network firmware needs inventory and end-of-life tracking, not just CVE scanning.

Part 6: targets by sector and geography

The sector pattern

Across the advisories, the emphasized sectors are government and diplomatic networks, think tanks and NGOs, energy and other critical infrastructure, defense and aerospace, transport and logistics, telecommunications, research and, for specific operations, sport and elections. Geography centers on Ukraine, NATO and EU members, the US, the UK and Canada, with Georgia, the Republic of Korea (Pyeongchang), Japan (Tokyo 2020), Afghanistan and India (Turla via Storm-0156), Poland and Czechia in specific operations (**Inferred** from [30] [51] [55] [40] [38] [15] [74] [58] [47]).

Service and unit	Sectors the sources name	Healthcare named?	Sources
GRU Unit 26165 (APT28)	Government, travel and hospitality, research institutions, NGOs (AA22-110A); defense industry, transport hubs, maritime, air traffic management, IT services in 12 European countries and the US (AA25-141A); sport and anti-doping	Not in AA22-110A or AA25-141A; HC3 (2022) lists healthcare among APT28 targets; HC3 (2024) says APT28 targeted pharmaceutical companies and clinical researchers for COVID-19 IP	[30] [55] [32] [50]
GRU Unit 74455 (Sandworm)	Electric power, government, media, Olympics; water, food and agriculture, energy through CARR	NotPetya collateral (HC3); not in AA25-343A	[60] [32] [57]
GRU Unit 29155	Government services, financial services, transportation, energy, healthcare	Yes (AA24-249A)	[51]
SVR (APT29)	Government, diplomatic, think tanks, healthcare, energy; IT companies; aviation, education, law enforcement, local councils, military	Yes (NCSC 2020, UK 2021, AA24-057A, HC3 2022 and 2024)	[12] [21] [45] [32] [50]
FSB Center 16 (Turla, Snake)	Education, small businesses, media, government facilities, financial services, critical manufacturing, communications (US victims)	Not in AA23-129A; HC3 (2022) lists research and pharmaceutical companies	[38] [32]
FSB Center 16 (Berserk Bear, Static Tundra)	Energy, nuclear, water, aviation, critical manufacturing; SLTT government	Yes: Healthcare and Public Health among the most at risk (AA26-194A)	[5] [16] [62]
FSB Center 18 (Star Blizzard)	Academia, defense, government, NGOs, think tanks, politicians; DOE facilities	Not in AA23-341A	[40]

The healthcare thread

The healthcare record is scattered across espionage advisories, sector notes and ransomware alerts, so it is gathered here in one place. It falls into four kinds of exposure.

1. State espionage against research and clinical data. The 2020 NCSC and CSE advisory says APT29 predominantly targets governmental, diplomatic, think-tank, healthcare and energy organisations, and that throughout 2020 it targeted organisations developing COVID-19 vaccines in Canada, the US and the UK, highly likely to steal vaccine development and testing IP (**Confirmed** [12] [13]). It used public exploits for Citrix, Pulse Secure, FortiGate and Zimbra, credential spear-phishing, and WellMess (in use since at least 2018) and WellMail, neither previously tied to APT29 in public (**Confirmed** [12]). Microsoft reported in November 2020 that three nation-state actors, one of them Strontium, which it says originates from Russia, targeted seven companies researching COVID-19 vaccines and treatments in Canada, France, India, South Korea and the US; Microsoft does not say which actor hit which company (**Confirmed** [67]). HC3's 2024 profile says both Midnight Blizzard and APT28 targeted foreign pharmaceutical companies and clinical researchers for vaccine and treatment IP, and adds Australia to the 2020 target countries, which the NCSC advisory does not list (**Confirmed** [50]; the two sources disagree). WADA's 2016 statement shows the same appetite for medical data in another setting: APT28 published athletes' confidential Therapeutic Use Exemptions (**Confirmed** [1]).

2. Supply-chain and software exposure. SVR exploitation of TeamCity in 2023 reached companies making software for billing and medical devices (**Confirmed** [42]). HC3's 2022 deck says SolarWinds affected a US hospital, but that claim rests in the deck on a press article, not a primary record; the SolarWinds study is the home for it (**Reported** [32]). HC3 also flags Midnight Blizzard's 2023 social engineering over Microsoft Teams, which it calls a staple in healthcare (**Confirmed** [50]).

3. Collateral damage and broad critical-infrastructure targeting. HC3's 2022 brief states that NotPetya shut down a US pharmaceutical manufacturer and affected the medical record systems of dozens of US hospitals (**Confirmed** [32]; the home study is SANDWORM_OPERATIONS.md). A press account of the 2023 New Jersey appellate ruling reports that NotPetya infected about 40,000 Merck computers in 64 countries and that the insurers' war exclusion did not bar Merck's claim (**Reported** [104]; the court opinion did not load). AA22-011A names Healthcare and Public Health among sectors Russian state actors have targeted, without naming an incident; AA24-249A lists healthcare among Unit 29155's target sectors; AA26-194A names Healthcare and Public Health among those most at risk from FSB Center 16's router campaign (**Confirmed** [26] [51] [62]). Void Blizzard and Storm-2372 both list healthcare or health among their targets (**Confirmed** [76] [75]).

4. Hacktivist and criminal disruption. HC3 records KillNet's reported threat against ventilators in British hospitals and the UK health ministry in May 2022, KillMilk's threat to sell Americans' health data and KillNet's targeting of Humana Military in December 2022, and more than 90 coordinated DDoS attacks on 28 January 2023, 55% against health systems with at least one hospital or hospitals with Level I trauma centres, across 48 US states (alleged) (**Confirmed** [33] [35]). HC3 advises treating KillNet's claims with caution. On the criminal side, AA20-302A warned on 28 October 2020 of an increased and imminent threat to US hospitals from TrickBot, BazarLoader, Ryuk and Conti, without attributing the operators to Russia (**Confirmed** [18]); Treasury later said Trickbot targeted hospitals during the pandemic, including three Minnesota facilities that had to divert ambulances (**Confirmed** [34]); the FBI counted at least 16 Conti attacks on US healthcare and first-responder networks in a year (**Confirmed** [23]); AA22-110A says Conti affiliates targeted US healthcare and first responders and that Emotet's operators targeted healthcare (**Confirmed** [30]); the HSE review in

Ireland traces Conti from a phishing email on 16 March 2021 to detonation on 14 May, after which all HSE IT was switched off and recovery took over four months, without stating the attackers' nationality (**Confirmed** [25]); AA24-131A says Black Basta hit healthcare among 12 of 16 critical infrastructure sectors and over 500 organisations, with no origin stated (**Confirmed** [49]); and Treasury's LockBit designation says affiliates attacked healthcare among other sectors (**Confirmed** [48]). HC3's 2024 profile groups LockBit, Royal, Black Basta and ALPHV among Russian actors that regularly attack the sector (**Confirmed** [50]). The ALPHV attack on Change Healthcare has its own study: Change Healthcare 2024.

What the record does not show. None of the government documents opened for this study attributes a disruptive attack on a hospital to a Russian state unit. The state record on healthcare is espionage, collateral damage and sector lists; the attacks that disrupted care came from criminal ransomware crews (Trickbot's ambulance diversions, Conti at the HSE), some of whom Treasury links to the intelligence services, while pro-Russia hacktivists added DDoS waves and threats (**Inferred** from [12] [32] [51] [62] [33] [34]). In ATT&CK's own descriptions of the Russia-linked groups, healthcare appears only twice: Turla lists pharmaceutical companies and Wizard Spider lists hospitals (**Inferred** from [93]).

Part 7: the defender view

What the advisories ask for

The advisories share most of their recommendations, which makes them a useful checklist across all three services.

Control area	What the advisories say	Sources
Patching	Patch KEV first; then critical internet-facing remote code execution; run regular automated vulnerability scans; organisations that did not patch TeamCity promptly should assume compromise	[30] [51] [42]
Identity	Enforce MFA, and phishing-resistant MFA on webmail and VPN (passkeys or PKI smartcards); strong unique passwords incl. service accounts; lockout at 5 to 10 attempts; disable dormant accounts; least privilege for service accounts; canary service accounts with alerting; short session lifetimes; restrict device enrollment; MFA alone does not stop session-cookie theft	[30] [51] [55] [45] [40]
Network devices	Disable Smart Install, SNMPv1 and v2c and Telnet; block TCP 4786 at the boundary; block outbound TFTP; keep management interfaces off the internet	[6]
Legacy protocols	Restrict NTLM and WDigest; block NTLM and SMB to external infrastructure; plan NTLM retirement; remove Group Policy Preference passwords; enable Credential Guard	[30] [55]
IT and OT	Segment IT and OT with a DMZ; watch for unexpected controller reboots	[30] [26]
Endpoint	EDR on mail servers and domain controllers first; attack surface reduction rules; PowerShell Constrained Language Mode; restrict and update PowerShell; detect Impacket, PsExec and WMI	[55] [51]
Cloud and mail	Audit app-only permissions and ApplicationImpersonation holders; review EWS full_access_as_app grants; disable or monitor mail forwarding rules; audit for privileged certificate access and rogue identity providers; require authentication for device registration	[69] [40] [22]

Control area	What the advisories say	Sources
Recovery	Offline, immutable, regularly tested backups; after a deep compromise, reset krbtgt twice and consider rebuilding the AD forest	[30] [16]
Validation	Test security controls against each advisory's ATT&CK technique list	[51] [42]

Detection and hunting leads in the primaries

- **Identity logs.** Failed logins across many accounts (spraying), impossible travel, activity on dormant accounts, atypical user agents, and access to `ntds.dit` on domain controllers (AA22-011A) (**Confirmed** [26]). AA24-057A prefers low-false-positive indicators such as user-agent changes over IP blocklists, because the SVR routes through residential proxies (**Confirmed** [45]).
- **EDR health.** Alert when EDR is disabled or loses contact, bring test and development systems under the security baseline, and hunt TeamCity logs for configuration changes; AA23-347A supplies Sigma rules and a GraphicalProton YARA rule (**Confirmed** [42]).
- **Kernel implants.** For Snake, AA23-129A rates memory analysis as the most effective detection because network and host signatures are easily changed (**Confirmed** [38]).
- **Microsoft 365.** Entra ID Protection password-spray detections, OAuth app permission audits and EWS grant reviews (**Confirmed** [69]).
- **OT.** Unexpected controller reboots (AA22-011A) and the ICS-protocol interest the FBI noted in Center 16's device collection (**Confirmed** [26] [56]).

A vulnerability- and exposure-management reading

The defender view for a VM program, as this study reads the record (**Inferred** unless a source is named):

1. **Edge first.** `T1190` and `T1133` are the common entry across GRU, SVR and FSB units, and every CVE these advisories name is in KEV. A remediation tier for "KEV and named in a Russia advisory" covering VPN, mail, collaboration, CI/CD and network devices is the single highest-yield control [26] [51] [55] [42] [56] [92].
2. **End-of-life gear is exposure, not just vulnerability.** The FSB device campaign used a 2018 CVE and legacy SNMP and Smart Install [56] [6]. Inventory end-of-life routers and switches as findings with owners and dates.
3. **Identity findings belong in the same backlog as CVEs.** "Account without MFA", "dormant account still enabled", "over-privileged service account" and "legacy OAuth app with app-only mail permissions" are the conditions the SVR exploited [45] [22] [69].
4. **Phishing-resistant MFA, not any MFA.** AA24-249A and AA25-141A ask for it, and AA23-341A shows why: cookie theft defeats ordinary MFA [51] [55] [40].
5. **Measure the gap between KEV and patch.** The 2018 Smart Install CVE shows that knowing is not the bottleneck [92] [56].

Where the library's countermeasure data is thin

The library's D3FEND mapping has no countermeasure rows for `T1059.001` PowerShell, `T1588.002` Obtain Capabilities: Tool, `T1595.002` Vulnerability Scanning, `T1098.005` Device Registration, `T1110` Brute Force (the parent), `T1485` Data Destruction, `T1550.002` Pass the Hash, `T1606.002` SAML Tokens or `T1685.005` Clear Windows Event Logs, and maps `T1133` External Remote Services only to Session Termination (**Confirmed** [102]). Several of these are among the most widely recorded techniques across the Russian groups, so a coverage tool driven by D3FEND alone will under-count controls against them. The live D3FEND page for Multi-factor Authentication labels its ATT&CK links as inferred and experimental, so D3FEND pairings in this study are the library's mapping, not an official pairing (**Confirmed** [91]).

Part 8: MITRE mappings

This part binds the verified record to MITRE ATT&CK v20.0, CAPEC, CWE and D3FEND. Every ID was checked against the live ATT&CK pages or the library's data files on 3 November 2026; IDs that could not be checked are left out. Revoked IDs that advisories still cite are shown under their v20 successors as `data/attack/superseded_by.jsonl` records them: `T1574.002` DLL Side-Loading becomes `T1574.001` DLL, `T1562.001` becomes `T1685` Disable or Modify Tools, and `T1070.001` becomes `T1685.005` Clear Windows Event Logs (**Confirmed** [100]).

Enterprise techniques

The table holds the union of three sets: the techniques ATT&CK v20.0 records at group level for the nine Russia-attributed groups with a government-named sponsor or a TsNIIKhM link (APT28, APT29, Sandworm Team, Turla, Dragonfly, Gamaredon Group, Ember Bear, Star Blizzard, TEMP.Veles); the techniques recorded only on their eight attributed campaigns; and the technique IDs stated in the ATT&CK tables of eight joint advisories (AA20-296A, AA22-011A, AA23-129A, AA23-341A, AA23-347A, AA24-057A, AA24-249A, AA25-141A). That is 330 live Enterprise techniques: 252 recorded at group level, 41 more only on campaigns and 37 more only in the advisories; 154 of the 330 are named in at least one advisory (**Confirmed** [100] [101]). Saint Bear, Winter Vivern, Wizard Spider and Indrik Spider are left out because no government source opened for this study names a Russian service for them, and campaign C0063 is left out because ATT&CK associates it with no group. One advisory list is partial: the 2021 GRU brute-force advisory's tables did not load (**Confirmed** [24]).

Counting each technique once per tactic it serves, the tally is Stealth 49, Persistence 42, Privilege Escalation 36, Command and Control 34, Credential Access 34, Discovery 31, Resource Development 29, Collection 24, Reconnaissance 23, Execution 21, Initial Access 19, Lateral Movement 18, Defense Impairment 14, Impact 10 and Exfiltration 8 (computed from [100]).

The most widely recorded techniques across the nine groups are `T1005` Data from Local System, `T1059.001` PowerShell and `T1588.002` Tool (seven groups each); `T1566.001` Spearphishing Attachment, `T1105` Ingress Tool Transfer, `T1204.002` Malicious File, `T1036.005` Match Legitimate Resource Name or Location and `T1070.004` File Deletion (six); and `T1190`, `T1078`, `T1133`, `T1595.002`, `T1505.003`, `T1053.005` and `T1203` (five) (**Confirmed** [101]). `T1190` is the ID stated in the most advisories, six of the eight (**Inferred** count over the advisories opened).

"Recorded by ATT&CK for" names the groups and campaigns in the library's copy of ATT&CK v20.0, which matched the live pages on the technique fields checked; the APT28 and Sandworm software lists differ slightly

from the live pages (Part 2) [96] [80] [82]. "Stated in advisory" gives the advisory and its date. Source numbers refer to the Sources table.

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1001	Data Obfuscation	Command and Control	Gamaredon Group	AA23-129A (2023-05-09)	[38] [85] [101]
T1001.001	Data Obfuscation: Junk Data	Command and Control	APT28	Not named in the advisories opened	[80] [101]
T1001.002	Data Obfuscation: Steganography	Command and Control	C0023 Operation Ghost (2013-09 to 2019-10)	Not named in the advisories opened	[94]
T1001.003	Data Obfuscation: Protocol or Service Impersonation	Command and Control	Not recorded for these groups or campaigns	AA23-129A (2023-05-09)	[38]
T1003	OS Credential Dumping	Credential Access	APT28; Ember Bear	AA23-129A (2023-05-09); AA23-347A (2023-12-12); AA24-249A (2024-09-05)	[38] [42] [51] [80] [86] [101]
T1003.001	OS Credential Dumping: LSASS Memory	Credential Access	APT28; Sandworm Team; Ember Bear; C0025 2016 Ukraine Electric Power Attack (2016-12); C0030 Triton SIS Attack (2017-06 to 2017-08); C0032 (the Triton IT foothold, 2014-10 to 2017-01)	AA23-347A (2023-12-12); AA24-249A (2024-09-05)	[42] [51] [80] [82] [86] [94] [101]
T1003.002	OS Credential Dumping: Security Account Manager	Credential Access	APT29; Dragonfly; Ember Bear; C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	AA23-347A (2023-12-12); AA24-249A (2024-09-05)	[42] [51] [81] [84] [86] [94] [97] [101]
T1003.003	OS Credential Dumping: NTDS	Credential Access	APT28; Sandworm Team; Dragonfly; C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	AA22-011A (2022-01-11); AA25-141A (2025-05)	[26] [55] [80] [82] [84] [94] [97] [101]
T1003.004	OS Credential Dumping: LSA Secrets	Credential Access	APT29; Dragonfly; Ember Bear	Not named in the advisories opened	[81] [84] [86] [101]
T1003.006	OS Credential Dumping: DCSync	Credential Access	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]
T1005	Data from Local System	Collection	APT28; Turla; APT29; Sandworm Team; Dragonfly; Gamaredon Group; Ember Bear; C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[80] [81] [82] [83] [84] [85] [86] [94] [101]
T1006	Direct Volume Access	Stealth	C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	Not named in the advisories opened	[94] [97]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1007	System Service Discovery	Discovery	Turla	Not named in the advisories opened	[83] [101]
T1012	Query Registry	Discovery	Turla; Dragonfly; Gamaredon Group	Not named in the advisories opened	[83] [84] [85] [101]
T1014	Rootkit	Stealth	APT28	AA23-129A (2023-05-09)	[38] [80] [101]
T1016	System Network Configuration Discovery	Discovery	Turla; Dragonfly	Not named in the advisories opened	[83] [84] [101]
T1016.001	System Network Configuration Discovery: Internet Connection Discovery	Discovery	Turla; APT29; Gamaredon Group; C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[81] [83] [85] [94] [101]
T1016.002	System Network Configuration Discovery: Wi-Fi Discovery	Discovery	C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	Not named in the advisories opened	[94] [97]
T1018	Remote System Discovery	Discovery	Turla; Sandworm Team; Dragonfly; Ember Bear; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0025 2016 Ukraine Electric Power Attack (2016-12); C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01)	Not named in the advisories opened	[82] [83] [84] [86] [94] [101]
T1020	Automated Exfiltration	Exfiltration	Gamaredon Group	AA23-347A (2023-12-12)	[42] [85] [101]
T1021	Remote Services	Lateral Movement	Ember Bear	Not named in the advisories opened	[86] [101]
T1021.001	Remote Services: Remote Desktop Protocol	Lateral Movement	Dragonfly; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0032 (the Triton IT foothold, 2014-10 to 2017-01); C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	AA25-141A (2025-05)	[55] [84] [94] [97] [101]
T1021.002	Remote Services: SMB/Windows Admin Shares	Lateral Movement	APT28; Turla; Sandworm Team; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0025 2016 Ukraine Electric Power Attack (2016-12); C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	Not named in the advisories opened	[80] [82] [83] [94] [97] [101]
T1021.004	Remote Services: SSH	Lateral Movement	C0032 (the Triton IT foothold, 2014-10 to 2017-01)	Not named in the advisories opened	[94]
T1021.005	Remote Services: VNC	Lateral Movement	Gamaredon Group	Not named in the advisories opened	[85] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1021.006	Remote Services: Windows Remote Management	Lateral Movement	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]
T1021.007	Remote Services: Cloud Services	Lateral Movement	APT29	Not named in the advisories opened	[81] [101]
T1025	Data from Removable Media	Collection	APT28; Turla; Gamaredon Group	Not named in the advisories opened	[80] [83] [85] [101]
T1027	Obfuscated Files or Information	Stealth	Sandworm Team; Gamaredon Group; C0025 2016 Ukraine Electric Power Attack (2016-12)	AA23-129A (2023-05-09)	[38] [82] [85] [94] [101]
T1027.001	Obfuscated Files or Information: Binary Padding	Stealth	APT29	AA23-347A (2023-12-12)	[42] [81] [101]
T1027.002	Obfuscated Files or Information: Software Packing	Stealth	APT29; C0025 2016 Ukraine Electric Power Attack (2016-12)	AA23-129A (2023-05-09)	[38] [81] [94] [101]
T1027.003	Obfuscated Files or Information: Steganography	Stealth	C0023 Operation Ghost (2013-09 to 2019-10)	Not named in the advisories opened	[94]
T1027.004	Obfuscated Files or Information: Compile After Delivery	Stealth	Gamaredon Group	Not named in the advisories opened	[85] [101]
T1027.005	Obfuscated Files or Information: Indicator Removal from Tools	Stealth	Turla; C0030 Triton SIS Attack (2017-06 to 2017-08)	Not named in the advisories opened	[83] [94] [101]
T1027.006	Obfuscated Files or Information: HTML Smuggling	Stealth	APT29	Not named in the advisories opened	[81] [101]
T1027.010	Obfuscated Files or Information: Command Obfuscation	Stealth	Turla; Sandworm Team; Gamaredon Group	Not named in the advisories opened	[82] [83] [85] [101]
T1027.011	Obfuscated Files or Information: Fileless Storage	Stealth	Turla	Not named in the advisories opened	[83] [101]
T1027.012	Obfuscated Files or Information: LNK Icon Smuggling	Stealth	Gamaredon Group	Not named in the advisories opened	[85] [101]
T1027.013	Obfuscated Files or Information: Encrypted/Encoded File	Stealth	APT28	Not named in the advisories opened	[80] [101]
T1027.015	Obfuscated Files or Information: Compression	Stealth	Gamaredon Group	Not named in the advisories opened	[85] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1027.016	Obfuscated Files or Information: Junk Code Insertion	Stealth	Gamaredon Group	Not named in the advisories opened	[85] [101]
T1029	Scheduled Transfer	Exfiltration	Not recorded for these groups or campaigns	AA25-141A (2025-05)	[55]
T1030	Data Transfer Size Limits	Exfiltration	APT28	Not named in the advisories opened	[80] [101]
T1033	System Owner/User Discovery	Discovery	Sandworm Team; Dragonfly; Gamaredon Group	AA23-347A (2023-12-12)	[42] [82] [84] [85] [101]
T1036	Masquerading	Stealth	APT28; Sandworm Team; Ember Bear	AA23-129A (2023-05-09); AA23-347A (2023-12-12)	[38] [42] [80] [82] [86] [101]
T1036.004	Masquerading: Masquerade Task or Service	Stealth	C0024 SolarWinds Compromise (2019-08 to 2021-01); C0034 2022 Ukraine Electric Power Attack (2022-06 to 2022-10)	Not named in the advisories opened	[94]
T1036.005	Masquerading: Match Legitimate Resource Name or Location	Stealth	APT28; Turla; APT29; Sandworm Team; Gamaredon Group; Ember Bear; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0025 2016 Ukraine Electric Power Attack (2016-12); C0030 Triton SIS Attack (2017-06 to 2017-08); C0032 (the Triton IT foothold, 2014-10 to 2017-01)	Not named in the advisories opened	[80] [81] [82] [83] [85] [86] [94] [101]
T1036.008	Masquerading: Masquerade File Type	Stealth	C0025 2016 Ukraine Electric Power Attack (2016-12)	Not named in the advisories opened	[94]
T1036.010	Masquerading: Masquerade Account Name	Stealth	Dragonfly; C0025 2016 Ukraine Electric Power Attack (2016-12)	Not named in the advisories opened	[84] [94] [101]
T1037	Boot or Logon Initialization Scripts	Persistence, Privilege Escalation	APT29	Not named in the advisories opened	[81] [101]
T1037.001	Boot or Logon Initialization Scripts: Logon Script (Windows)	Persistence, Privilege Escalation	APT28	Not named in the advisories opened	[80] [101]
T1037.004	Boot or Logon Initialization Scripts: RC Scripts	Persistence, Privilege Escalation	APT29	Not named in the advisories opened	[81] [101]
T1039	Data from Network Shared Drive	Collection	APT28; Gamaredon Group	Not named in the advisories opened	[80] [85] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1040	Network Sniffing	Credential Access, Discovery	APT28; Sandworm Team; C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01)	AA23-129A (2023-05-09)	[38] [80] [82] [94] [101]
T1041	Exfiltration Over C2 Channel	Exfiltration	Sandworm Team; Gamaredon Group	AA23-347A (2023-12-12)	[42] [82] [85] [101]
T1046	Network Service Discovery	Discovery	Ember Bear	AA23-129A (2023-05-09); AA23-347A (2023-12-12); AA24-249A (2024-09-05)	[38] [42] [51] [86] [101]
T1047	Windows Management Instrumentation	Execution	APT29; Sandworm Team; Gamaredon Group; Ember Bear; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0025 2016 Ukraine Electric Power Attack (2016-12)	AA23-347A (2023-12-12)	[42] [81] [82] [85] [86] [94] [101]
T1048	Exfiltration Over Alternative Protocol	Exfiltration	Not recorded for these groups or campaigns	AA25-141A (2025-05)	[55]
T1048.002	Exfiltration Over Alternative Protocol: Exfiltration Over Asymmetric Encrypted Non-C2 Protocol	Exfiltration	APT28; C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[80] [94] [101]
T1049	System Network Connections Discovery	Discovery	Turla; Sandworm Team	AA23-347A (2023-12-12)	[42] [82] [83] [101]
T1053.005	Scheduled Task/Job: Scheduled Task	Execution, Persistence, Privilege Escalation	APT29; Sandworm Team; Dragonfly; Gamaredon Group; Ember Bear; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0030 Triton SIS Attack (2017-06 to 2017-08); C0032 (the Triton IT foothold, 2014-10 to 2017-01); C0034 2022 Ukraine Electric Power Attack (2022-06 to 2022-10)	AA23-347A (2023-12-12); AA25-141A (2025-05)	[42] [55] [81] [82] [84] [85] [86] [94] [101]
T1055	Process Injection	Stealth, Privilege Escalation	Turla; Gamaredon Group; C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01)	AA23-347A (2023-12-12)	[42] [83] [85] [94] [101]
T1055.001	Process Injection: Dynamic-link Library Injection	Stealth, Privilege Escalation	Turla	AA23-129A (2023-05-09)	[38] [83] [101]
T1056	Input Capture	Collection, Credential Access	Not recorded for these groups or campaigns	AA25-141A (2025-05)	[55]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1056.001	Input Capture: Keylogging	Collection, Credential Access	APT28; Sandworm Team; C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01)	AA23-129A (2023-05-09)	[38] [80] [82] [94] [101]
T1056.003	Input Capture: Web Portal Capture	Collection, Credential Access	C0030 Triton SIS Attack (2017-06 to 2017-08)	Not named in the advisories opened	[94]
T1057	Process Discovery	Discovery	APT28; Turla; Gamaredon Group; C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA23-347A (2023-12-12)	[42] [80] [83] [85] [94] [101]
T1059	Command and Scripting Interpreter	Execution	Dragonfly	AA25-141A (2025-05)	[55] [84] [101]
T1059.001	Command and Scripting Interpreter: PowerShell	Execution	APT28; Turla; APT29; Sandworm Team; Dragonfly; Gamaredon Group; Ember Bear; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0025 2016 Ukraine Electric Power Attack (2016-12); C0030 Triton SIS Attack (2017-06 to 2017-08); C0032 (the Triton IT foothold, 2014-10 to 2017-01); C0034 2022 Ukraine Electric Power Attack (2022-06 to 2022-10); C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	AA23-129A (2023-05-09); AA23-347A (2023-12-12); AA24-249A (2024-09-05); AA25-141A (2025-05)	[38] [42] [51] [55] [80] [81] [82] [83] [84] [85] [86] [94] [97] [101]
T1059.003	Command and Scripting Interpreter: Windows Command Shell	Execution	APT28; Turla; Dragonfly; Gamaredon Group; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0025 2016 Ukraine Electric Power Attack (2016-12); C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	AA22-011A (2022-01-11); AA23-347A (2023-12-12); AA25-141A (2025-05)	[26] [42] [55] [80] [83] [84] [85] [94] [97] [101]
T1059.005	Command and Scripting Interpreter: Visual Basic	Execution	Turla; Sandworm Team; Gamaredon Group; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0025 2016 Ukraine Electric Power Attack (2016-12); C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01)	AA25-141A (2025-05)	[55] [82] [83] [85] [94] [101]
T1059.006	Command and Scripting Interpreter: Python	Execution	Turla; APT29; Dragonfly	AA25-141A (2025-05)	[55] [81] [83] [84] [101]
T1059.007	Command and Scripting Interpreter: JavaScript	Execution	Turla; Star Blizzard	Not named in the advisories opened	[83] [87] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1059.009	Command and Scripting Interpreter: Cloud API	Execution	APT29	Not named in the advisories opened	[81] [101]
T1068	Exploitation for Privilege Escalation	Privilege Escalation	APT28; Turla; APT29	AA23-347A (2023-12-12)	[42] [80] [81] [83] [101]
T1069	Permission Groups Discovery	Discovery	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]
T1069.001	Permission Groups Discovery: Local Groups	Discovery	Turla	Not named in the advisories opened	[83] [101]
T1069.002	Permission Groups Discovery: Domain Groups	Discovery	Turla; Dragonfly; C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[83] [84] [94] [101]
T1070	Indicator Removal	Stealth	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]
T1070.004	Indicator Removal: File Deletion	Stealth	APT28; APT29; Sandworm Team; Dragonfly; Gamaredon Group; Ember Bear; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01); C0032 (the Triton IT foothold, 2014-10 to 2017-01)	Not named in the advisories opened	[80] [81] [82] [84] [85] [86] [94] [101]
T1070.006	Indicator Removal: Timestomp	Stealth	APT28; APT29; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0032 (the Triton IT foothold, 2014-10 to 2017-01)	Not named in the advisories opened	[80] [81] [94] [101]
T1070.008	Indicator Removal: Clear Mailbox Data	Stealth	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]
T1071	Application Layer Protocol	Command and Control	Not recorded for these groups or campaigns	AA23-129A (2023-05-09)	[38]
T1071.001	Application Layer Protocol: Web Protocols	Command and Control	APT28; Turla; Sandworm Team; Gamaredon Group; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01)	AA23-129A (2023-05-09); AA24-249A (2024-09-05)	[38] [51] [80] [82] [83] [85] [94] [101]
T1071.002	Application Layer Protocol: File Transfer Protocols	Command and Control	Dragonfly	Not named in the advisories opened	[84] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1071.003	Application Layer Protocol: Mail Protocols	Command and Control	APT28; Turla	AA23-129A (2023-05-09)	[38] [80] [83] [101]
T1071.004	Application Layer Protocol: DNS	Command and Control	Ember Bear	AA23-129A (2023-05-09); AA24-249A (2024-09-05)	[38] [51] [86] [101]
T1072	Software Deployment Tools	Execution, Lateral Movement	Sandworm Team	Not named in the advisories opened	[82] [101]
T1074	Data Staged	Collection	Not recorded for these groups or campaigns	AA23-129A (2023-05-09)	[38]
T1074.001	Data Staged: Local Data Staging	Collection	APT28; Dragonfly; C0032 (the Triton IT foothold, 2014-10 to 2017-01); C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	Not named in the advisories opened	[80] [84] [94] [97] [101]
T1074.002	Data Staged: Remote Data Staging	Collection	APT28; C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[80] [94] [101]
T1078	Valid Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	APT28; APT29; Sandworm Team; Dragonfly; Star Blizzard; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01); C0032 (the Triton IT foothold, 2014-10 to 2017-01)	AA20-296A (2020-10-22); AA22-011A (2022-01-11); AA23-129A (2023-05-09); AA23-341A (2023-12-07)	[16] [26] [38] [40] [80] [81] [82] [84] [87] [94] [101]
T1078.001	Valid Accounts: Default Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	Ember Bear	AA24-249A (2024-09-05)	[51] [86] [101]
T1078.002	Valid Accounts: Domain Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	Sandworm Team; C0023 Operation Ghost (2013-09 to 2019-10); C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[82] [94] [101]
T1078.003	Valid Accounts: Local Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	Turla; APT29; C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[81] [83] [94] [101]
T1078.004	Valid Accounts: Cloud Accounts	Stealth, Persistence, Privilege Escalation, Initial Access	APT28; APT29; C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA24-057A (2024-02-26)	[45] [80] [81] [94] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1080	Taint Shared Content	Lateral Movement	Gamaredon Group	Not named in the advisories opened	[85] [101]
T1082	System Information Discovery	Discovery	Turla; Sandworm Team; Gamaredon Group	Not named in the advisories opened	[82] [83] [85] [101]
T1083	File and Directory Discovery	Discovery	APT28; Turla; Sandworm Team; Dragonfly; Gamaredon Group; C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA23-129A (2023-05-09)	[38] [80] [82] [83] [84] [85] [94] [101]
T1087	Account Discovery	Discovery	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]
T1087.001	Account Discovery: Local Account	Discovery	Turla	Not named in the advisories opened	[83] [101]
T1087.002	Account Discovery: Domain Account	Discovery	Turla; Sandworm Team; Dragonfly; C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA25-141A (2025-05)	[55] [82] [83] [84] [94] [101]
T1087.003	Account Discovery: Email Account	Discovery	Sandworm Team	Not named in the advisories opened	[82] [101]
T1087.004	Account Discovery: Cloud Account	Discovery	APT29	Not named in the advisories opened	[81] [101]
T1090	Proxy	Command and Control	Turla; Sandworm Team; Gamaredon Group	Not named in the advisories opened	[82] [83] [85] [101]
T1090.001	Proxy: Internal Proxy	Command and Control	Turla; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	Not named in the advisories opened	[83] [94] [97] [101]
T1090.002	Proxy: External Proxy	Command and Control	APT28; APT29	AA24-057A (2024-02-26); AA25-141A (2025-05)	[45] [55] [80] [81] [101]
T1090.003	Proxy: Multi-hop Proxy	Command and Control	APT28; APT29; Gamaredon Group; Ember Bear	AA22-011A (2022-01-11); AA23-129A (2023-05-09); AA24-249A (2024-09-05); AA25-141A (2025-05)	[26] [38] [51] [55] [80] [81] [85] [86] [101]
T1090.004	Proxy: Domain Fronting	Command and Control	APT29	Not named in the advisories opened	[81] [101]
T1091	Replication Through Removable Media	Lateral Movement, Initial Access	APT28; Gamaredon Group	Not named in the advisories opened	[80] [85] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1092	Communication Through Removable Media	Command and Control	APT28	Not named in the advisories opened	[80] [101]
T1095	Non-Application Layer Protocol	Command and Control	Gamaredon Group; Ember Bear; C0034 2022 Ukraine Electric Power Attack (2022-06 to 2022-10)	AA23-129A (2023-05-09); AA24-249A (2024-09-05)	[38] [51] [85] [86] [94] [101]
T1098	Account Manipulation	Persistence, Privilege Escalation	C0025 2016 Ukraine Electric Power Attack (2016-12)	AA23-347A (2023-12-12)	[42] [94]
T1098.001	Account Manipulation: Additional Cloud Credentials	Persistence, Privilege Escalation	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]
T1098.002	Account Manipulation: Additional Email Delegate Permissions	Persistence, Privilege Escalation	APT28; APT29; C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA25-141A (2025-05)	[55] [80] [81] [94] [101]
T1098.003	Account Manipulation: Additional Cloud Roles	Persistence, Privilege Escalation	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]
T1098.005	Account Manipulation: Device Registration	Persistence, Privilege Escalation	APT29; C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA24-057A (2024-02-26)	[45] [81] [94] [101]
T1098.007	Account Manipulation: Additional Local or Domain Groups	Persistence, Privilege Escalation	Dragonfly	Not named in the advisories opened	[84] [101]
T1102	Web Service	Command and Control	Turla; Gamaredon Group	Not named in the advisories opened	[83] [85] [101]
T1102.002	Web Service: Bidirectional Communication	Command and Control	APT28; Turla; Sandworm Team; Gamaredon Group; C0023 Operation Ghost (2013-09 to 2019-10)	Not named in the advisories opened	[80] [82] [83] [85] [94] [101]
T1102.003	Web Service: One-Way Communication	Command and Control	Gamaredon Group	Not named in the advisories opened	[85] [101]
T1104	Multi-Stage Channels	Command and Control	Not recorded for these groups or campaigns	AA23-129A (2023-05-09); AA25-141A (2025-05)	[38] [55]
T1105	Ingress Tool Transfer	Command and Control	APT28; Turla; APT29; Sandworm Team; Dragonfly; Gamaredon Group; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01)	AA24-249A (2024-09-05)	[51] [80] [81] [82] [83] [84] [85] [94] [101]
T1106	Native API	Execution	Turla; Sandworm Team; Gamaredon Group	AA23-129A (2023-05-09)	[38] [82] [83] [85] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1110	Brute Force	Credential Access	APT28; Turla; Dragonfly; Ember Bear; C0025 2016 Ukraine Electric Power Attack (2016-12)	AA20-296A (2020-10-22); AA24-057A (2024-02-26); AA25-141A (2025-05)	[16] [45] [55] [80] [83] [84] [86] [94] [101]
T1110.001	Brute Force: Password Guessing	Credential Access	APT28; APT29	AA22-011A (2022-01-11); AA25-141A (2025-05)	[26] [55] [80] [81] [101]
T1110.002	Brute Force: Password Cracking	Credential Access	Dragonfly	Not named in the advisories opened	[84] [101]
T1110.003	Brute Force: Password Spraying	Credential Access	APT28; APT29; Ember Bear; C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	AA22-011A (2022-01-11); AA24-249A (2024-09-05); AA25-141A (2025-05)	[26] [51] [55] [80] [81] [86] [94] [97] [101]
T1111	Multi-Factor Authentication Interception	Credential Access	Not recorded for these groups or campaigns	AA25-141A (2025-05)	[55]
T1112	Modify Registry	Defense Impairment, Persistence	Turla; Dragonfly; Gamaredon Group; Ember Bear; C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01)	AA23-129A (2023-05-09)	[38] [83] [84] [85] [86] [94] [101]
T1113	Screen Capture	Collection	APT28; Dragonfly; Gamaredon Group	Not named in the advisories opened	[80] [84] [85] [101]
T1114	Email Collection	Collection	Ember Bear	AA24-249A (2024-09-05); AA25-141A (2025-05)	[51] [55] [86] [101]
T1114.002	Email Collection: Remote Email Collection	Collection	APT28; APT29; Dragonfly; Star Blizzard; C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA23-341A (2023-12-07); AA25-141A (2025-05)	[40] [55] [80] [81] [84] [87] [94] [101]
T1114.003	Email Collection: Email Forwarding Rule	Collection	Star Blizzard	AA23-341A (2023-12-07)	[40] [87] [101]
T1119	Automated Collection	Collection	APT28; Gamaredon Group; Ember Bear	AA23-129A (2023-05-09); AA25-141A (2025-05)	[38] [55] [80] [85] [86] [101]
T1120	Peripheral Device Discovery	Discovery	APT28; Turla; Gamaredon Group	Not named in the advisories opened	[80] [83] [85] [101]
T1124	System Time Discovery	Discovery	Turla	Not named in the advisories opened	[83] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1125	Video Capture	Collection	Ember Bear	AA24-249A (2024-09-05); AA25-141A (2025-05)	[51] [55] [86] [101]
T1132	Data Encoding	Command and Control	Not recorded for these groups or campaigns	AA23-129A (2023-05-09)	[38]
T1132.001	Data Encoding: Standard Encoding	Command and Control	Sandworm Team	Not named in the advisories opened	[82] [101]
T1132.002	Data Encoding: Non-Standard Encoding	Command and Control	Not recorded for these groups or campaigns	AA23-129A (2023-05-09)	[38]
T1133	External Remote Services	Persistence, Initial Access	APT28; APT29; Sandworm Team; Dragonfly; Ember Bear; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01); C0032 (the Triton IT foothold, 2014-10 to 2017-01)	AA20-296A (2020-10-22); AA25-141A (2025-05)	[16] [55] [80] [81] [82] [84] [86] [94] [101]
T1134.001	Access Token Manipulation: Token Impersonation/Theft	Stealth, Privilege Escalation	APT28	Not named in the advisories opened	[80] [101]
T1134.002	Access Token Manipulation: Create Process with Token	Stealth, Privilege Escalation	Turla	Not named in the advisories opened	[83] [101]
T1135	Network Share Discovery	Discovery	Dragonfly	AA23-129A (2023-05-09)	[38] [84] [101]
T1136	Create Account	Persistence	C0025 2016 Ukraine Electric Power Attack (2016-12)	Not named in the advisories opened	[94]
T1136.001	Create Account: Local Account	Persistence	Dragonfly	Not named in the advisories opened	[84] [101]
T1136.002	Create Account: Domain Account	Persistence	C0025 2016 Ukraine Electric Power Attack (2016-12); C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01)	Not named in the advisories opened	[94]
T1136.003	Create Account: Cloud Account	Persistence	APT29	Not named in the advisories opened	[81] [101]
T1137	Office Application Startup	Persistence	Gamaredon Group	Not named in the advisories opened	[85] [101]
T1137.002	Office Application Startup: Office Test	Persistence	APT28	Not named in the advisories opened	[80] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1140	Deobfuscate/Decode Files or Information	Stealth	APT28; Turla; Sandworm Team; Gamaredon Group; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	AA23-129A (2023-05-09)	[38] [80] [82] [83] [85] [94] [97] [101]
T1187	Forced Authentication	Credential Access	Dragonfly	AA25-141A (2025-05)	[55] [84] [101]
T1189	Drive-by Compromise	Initial Access	APT28; Turla; Dragonfly	AA20-296A (2020-10-22)	[16] [80] [83] [84] [101]
T1190	Exploit Public-Facing Application	Initial Access	APT28; APT29; Sandworm Team; Dragonfly; Ember Bear; C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA20-296A (2020-10-22); AA22-011A (2022-01-11); AA23-129A (2023-05-09); AA23-347A (2023-12-12); AA24-249A (2024-09-05); AA25-141A (2025-05)	[16] [26] [38] [42] [51] [55] [80] [81] [82] [84] [86] [94] [101]
T1195	Supply Chain Compromise	Initial Access	Sandworm Team; Ember Bear	Not named in the advisories opened	[82] [86] [101]
T1195.002	Supply Chain Compromise: Compromise Software Supply Chain	Initial Access	Sandworm Team; Dragonfly; C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA22-011A (2022-01-11)	[26] [82] [84] [94] [101]
T1199	Trusted Relationship	Initial Access	APT28; APT29; Sandworm Team; C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA25-141A (2025-05)	[55] [80] [81] [82] [94] [101]
T1201	Password Policy Discovery	Discovery	Turla	Not named in the advisories opened	[83] [101]
T1203	Exploitation for Client Execution	Execution	APT28; APT29; Sandworm Team; Dragonfly; Ember Bear	AA23-347A (2023-12-12)	[42] [80] [81] [82] [84] [86] [101]
T1204.001	User Execution: Malicious Link	Execution	APT28; Turla; APT29; Sandworm Team; Gamaredon Group	AA25-141A (2025-05)	[55] [80] [81] [82] [83] [85] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1204.002	User Execution: Malicious File	Execution	APT28; APT29; Sandworm Team; Dragonfly; Gamaredon Group; Star Blizzard; C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01)	AA25-141A (2025-05)	[55] [80] [81] [82] [84] [85] [87] [94] [101]
T1210	Exploitation of Remote Services	Lateral Movement	APT28; Dragonfly; Ember Bear	AA23-347A (2023-12-12)	[42] [80] [84] [86] [101]
T1211	Exploitation for Stealth	Stealth	APT28	Not named in the advisories opened	[80] [101]
T1212	Exploitation for Credential Access	Credential Access	Not recorded for these groups or campaigns	AA22-011A (2022-01-11)	[26]
T1213	Data from Information Repositories	Collection	APT28; C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[80] [94] [101]
T1213.001	Data from Information Repositories: Confluence	Collection	Not recorded for these groups or campaigns	AA24-249A (2024-09-05)	[51]
T1213.002	Data from Information Repositories: Sharepoint	Collection	APT28	Not named in the advisories opened	[80] [101]
T1213.003	Data from Information Repositories: Code Repositories	Collection	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]
T1213.006	Data from Information Repositories: Databases	Collection	Turla; Sandworm Team	Not named in the advisories opened	[82] [83] [101]
T1218.005	System Binary Proxy Execution: Mshta	Stealth	APT29; Gamaredon Group	Not named in the advisories opened	[81] [85] [101]
T1218.011	System Binary Proxy Execution: Rundll32	Stealth	APT28; Sandworm Team; Gamaredon Group; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01)	Not named in the advisories opened	[80] [82] [85] [94] [101]
T1219	Remote Access Tools	Command and Control	Sandworm Team	Not named in the advisories opened	[82] [101]
T1221	Template Injection	Stealth	APT28; Dragonfly; Gamaredon Group	Not named in the advisories opened	[80] [84] [85] [101]
T1480	Execution Guardrails	Stealth	Gamaredon Group	Not named in the advisories opened	[85] [101]
T1482	Domain Trust Discovery	Discovery	C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA23-129A (2023-05-09)	[38] [94]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1484.001	Domain or Tenant Policy Modification: Group Policy Modification	Defense Impairment, Privilege Escalation	C0034 2022 Ukraine Electric Power Attack (2022-06 to 2022-10)	Not named in the advisories opened	[94]
T1484.002	Domain or Tenant Policy Modification: Trust Modification	Defense Impairment, Privilege Escalation	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]
T1485	Data Destruction	Impact	Sandworm Team; C0034 2022 Ukraine Electric Power Attack (2022-06 to 2022-10)	AA24-249A (2024-09-05)	[51] [82] [94] [101]
T1486	Data Encrypted for Impact	Impact	Sandworm Team	Not named in the advisories opened	[82] [101]
T1489	Service Stop	Impact	Sandworm Team	Not named in the advisories opened	[82] [101]
T1490	Inhibit System Recovery	Impact	Sandworm Team	Not named in the advisories opened	[82] [101]
T1491.001	Defacement: Internal Defacement	Impact	Gamaredon Group	Not named in the advisories opened	[85] [101]
T1491.002	Defacement: External Defacement	Impact	Sandworm Team; Ember Bear	Not named in the advisories opened	[82] [86] [101]
T1497.001	Virtualization/Sandbox Evasion: System Checks	Stealth, Discovery	Gamaredon Group	Not named in the advisories opened	[85] [101]
T1498	Network Denial of Service	Impact	APT28	Not named in the advisories opened	[80] [101]
T1499	Endpoint Denial of Service	Impact	Sandworm Team	Not named in the advisories opened	[82] [101]
T1505.001	Server Software Component: SQL Stored Procedures	Persistence	C0025 2016 Ukraine Electric Power Attack (2016-12)	AA23-347A (2023-12-12)	[42] [94]
T1505.003	Server Software Component: Web Shell	Persistence	APT28; APT29; Sandworm Team; Dragonfly; Ember Bear; C0032 (the Triton IT foothold, 2014-10 to 2017-01); C0034 2022 Ukraine Electric Power Attack (2022-06 to 2022-10)	AA24-249A (2024-09-05)	[51] [80] [81] [82] [84] [86] [94] [101]
T1518.001	Software Discovery: Security Software Discovery	Discovery	Turla; Gamaredon Group	Not named in the advisories opened	[83] [85] [101]
T1528	Steal Application Access Token	Credential Access	APT28; APT29	AA24-057A (2024-02-26)	[45] [80] [81] [101]
T1534	Internal Spearphishing	Lateral Movement	Gamaredon Group	Not named in the advisories opened	[85] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1539	Steal Web Session Cookie	Credential Access	Sandworm Team; Star Blizzard; C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA23-341A (2023-12-07)	[40] [82] [87] [94] [101]
T1542.003	Pre-OS Boot: Bootkit	Stealth, Persistence	APT28	Not named in the advisories opened	[80] [101]
T1543.002	Create or Modify System Process: Systemd Service	Persistence, Privilege Escalation	C0034 2022 Ukraine Electric Power Attack (2022-06 to 2022-10)	Not named in the advisories opened	[94]
T1543.003	Create or Modify System Process: Windows Service	Persistence, Privilege Escalation	C0025 2016 Ukraine Electric Power Attack (2016-12)	Not named in the advisories opened	[94]
T1546.003	Event Triggered Execution: Windows Management Instrumentation Event Subscription	Privilege Escalation, Persistence	Turla; APT29; C0023 Operation Ghost (2013-09 to 2019-10); C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[81] [83] [94] [101]
T1546.008	Event Triggered Execution: Accessibility Features	Privilege Escalation, Persistence	APT29	Not named in the advisories opened	[81] [101]
T1546.012	Event Triggered Execution: Image File Execution Options Injection	Privilege Escalation, Persistence	C0032 (the Triton IT foothold, 2014-10 to 2017-01)	Not named in the advisories opened	[94]
T1546.013	Event Triggered Execution: PowerShell Profile	Privilege Escalation, Persistence	Turla	Not named in the advisories opened	[83] [101]
T1546.015	Event Triggered Execution: Component Object Model Hijacking	Privilege Escalation, Persistence	APT28	Not named in the advisories opened	[80] [101]
T1546.016	Event Triggered Execution: Installer Packages	Privilege Escalation, Persistence	Not recorded for these groups or campaigns	AA23-129A (2023-05-09)	[38]
T1547	Boot or Logon Autostart Execution	Persistence, Privilege Escalation	Not recorded for these groups or campaigns	AA23-347A (2023-12-12)	[42]
T1547.001	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder	Persistence, Privilege Escalation	APT28; Turla; APT29; Dragonfly; Gamaredon Group	AA25-141A (2025-05)	[55] [80] [81] [83] [84] [85] [101]
T1547.004	Boot or Logon Autostart Execution: Winlogon Helper DLL	Persistence, Privilege Escalation	Turla	Not named in the advisories opened	[83] [101]
T1547.006	Boot or Logon Autostart Execution: Kernel Modules and Extensions	Persistence, Privilege Escalation	Not recorded for these groups or campaigns	AA23-129A (2023-05-09)	[38]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1547.009	Boot or Logon Autostart Execution: Shortcut Modification	Persistence, Privilege Escalation	Not recorded for these groups or campaigns	AA25-141A (2025-05)	[55]
T1548.002	Abuse Elevation Control Mechanism: Bypass User Account Control	Privilege Escalation	APT29	Not named in the advisories opened	[81] [101]
T1550	Use Alternate Authentication Material	Lateral Movement	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]
T1550.001	Use Alternate Authentication Material: Application Access Token	Lateral Movement	APT28; C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[80] [94] [101]
T1550.002	Use Alternate Authentication Material: Pass the Hash	Lateral Movement	APT28; Ember Bear	AA24-249A (2024-09-05)	[51] [80] [86] [101]
T1550.003	Use Alternate Authentication Material: Pass the Ticket	Lateral Movement	APT29	Not named in the advisories opened	[81] [101]
T1550.004	Use Alternate Authentication Material: Web Session Cookie	Lateral Movement	Star Blizzard; C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA23-341A (2023-12-07)	[40] [87] [94] [101]
T1552.001	Unsecured Credentials: Credentials In Files	Credential Access	Ember Bear	AA24-249A (2024-09-05)	[51] [86] [101]
T1552.004	Unsecured Credentials: Private Keys	Credential Access	C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA22-011A (2022-01-11)	[26] [94]
T1552.006	Unsecured Credentials: Group Policy Preferences	Credential Access	Not recorded for these groups or campaigns	AA25-141A (2025-05)	[55]
T1553.002	Subvert Trust Controls: Code Signing	Defense Impairment	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]
T1553.005	Subvert Trust Controls: Mark-of-the-Web Bypass	Defense Impairment	APT29	Not named in the advisories opened	[81] [101]
T1553.006	Subvert Trust Controls: Code Signing Policy Modification	Defense Impairment	Turla	Not named in the advisories opened	[83] [101]
T1554	Compromise Host Software Binary	Persistence	C0025 2016 Ukraine Electric Power Attack (2016-12)	Not named in the advisories opened	[94]
T1555	Credentials from Password Stores	Credential Access	C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA22-011A (2022-01-11)	[26] [94]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1555.003	Credentials from Password Stores: Credentials from Web Browsers	Credential Access	Sandworm Team; C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA23-347A (2023-12-12)	[42] [82] [94] [101]
T1555.004	Credentials from Password Stores: Windows Credential Manager	Credential Access	Turla	Not named in the advisories opened	[83] [101]
T1556.006	Modify Authentication Process: Multi-Factor Authentication	Defense Impairment, Persistence, Credential Access	Not recorded for these groups or campaigns	AA25-141A (2025-05)	[55]
T1556.007	Modify Authentication Process: Hybrid Identity	Defense Impairment, Persistence, Credential Access	APT29	Not named in the advisories opened	[81] [101]
T1557.004	Adversary-in-the-Middle : Evil Twin	Credential Access, Collection	APT28	Not named in the advisories opened	[80] [101]
T1558.001	Steal or Forge Kerberos Tickets: Golden Ticket	Credential Access	Not recorded for these groups or campaigns	AA23-347A (2023-12-12)	[42]
T1558.003	Steal or Forge Kerberos Tickets: Kerberoasting	Credential Access	C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA22-011A (2022-01-11)	[26] [94]
T1559	Inter-Process Communication	Execution	Not recorded for these groups or campaigns	AA23-129A (2023-05-09)	[38]
T1559.001	Inter-Process Communication: Component Object Model	Execution	Gamaredon Group	Not named in the advisories opened	[85] [101]
T1559.002	Inter-Process Communication: Dynamic Data Exchange	Execution	APT28	Not named in the advisories opened	[80] [101]
T1560	Archive Collected Data	Collection	APT28; Dragonfly; Ember Bear	AA24-249A (2024-09-05); AA25-141A (2025-05)	[51] [55] [80] [84] [86] [101]
T1560.001	Archive Collected Data: Archive via Utility	Collection	APT28; Turla; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	AA25-141A (2025-05)	[55] [80] [83] [94] [97] [101]
T1560.003	Archive Collected Data: Archive via Custom Method	Collection	Not recorded for these groups or campaigns	AA23-129A (2023-05-09)	[38]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1561.001	Disk Wipe: Disk Content Wipe	Impact	Gamaredon Group; C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	Not named in the advisories opened	[85] [94] [97] [101]
T1561.002	Disk Wipe: Disk Structure Wipe	Impact	Sandworm Team; Ember Bear	Not named in the advisories opened	[82] [86] [101]
T1564	Hide Artifacts	Stealth	Not recorded for these groups or campaigns	AA23-129A (2023-05-09); AA23-347A (2023-12-12)	[38] [42]
T1564.001	Hide Artifacts: Hidden Files and Directories	Stealth	APT28	AA23-347A (2023-12-12)	[42] [80] [101]
T1564.002	Hide Artifacts: Hidden Users	Stealth	Dragonfly	Not named in the advisories opened	[84] [101]
T1564.003	Hide Artifacts: Hidden Window	Stealth	APT28; Gamaredon Group	Not named in the advisories opened	[80] [85] [101]
T1564.012	Hide Artifacts: File/Path Exclusions	Stealth	Turla	Not named in the advisories opened	[83] [101]
T1566	Phishing	Initial Access	Not recorded for these groups or campaigns	AA25-141A (2025-05)	[55]
T1566.001	Phishing: Spearphishing Attachment	Initial Access	APT28; APT29; Sandworm Team; Dragonfly; Gamaredon Group; Star Blizzard; C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01)	AA23-341A (2023-12-07); AA25-141A (2025-05)	[40] [55] [80] [81] [82] [84] [85] [87] [94] [101]
T1566.002	Phishing: Spearphishing Link	Initial Access	Turla; APT29; Sandworm Team	AA23-341A (2023-12-07); AA25-141A (2025-05)	[40] [55] [81] [82] [83] [101]
T1566.003	Phishing: Spearphishing via Service	Initial Access	APT29	Not named in the advisories opened	[81] [101]
T1566.004	Phishing: Spearphishing Voice	Initial Access	Not recorded for these groups or campaigns	AA25-141A (2025-05)	[55]
T1567	Exfiltration Over Web Service	Exfiltration	APT28; C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	AA23-347A (2023-12-12)	[42] [80] [94] [97] [101]
T1567.002	Exfiltration Over Web Service: Exfiltration to Cloud Storage	Exfiltration	Turla; Ember Bear	AA24-249A (2024-09-05)	[51] [83] [86] [101]
T1568	Dynamic Resolution	Command and Control	APT29; Gamaredon Group; C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA23-347A (2023-12-12)	[42] [81] [85] [94] [101]
T1568.001	Dynamic Resolution: Fast Flux DNS	Command and Control	Gamaredon Group	Not named in the advisories opened	[85] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1569.002	System Services: Service Execution	Execution	Not recorded for these groups or campaigns	AA23-129A (2023-05-09)	[38]
T1570	Lateral Tool Transfer	Lateral Movement	Turla; Sandworm Team; Ember Bear; C0025 2016 Ukraine Electric Power Attack (2016-12); C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01); C0034 2022 Ukraine Electric Power Attack (2022-06 to 2022-10)	AA23-129A (2023-05-09)	[38] [82] [83] [86] [94] [101]
T1571	Non-Standard Port	Command and Control	Sandworm Team; Gamaredon Group; Ember Bear; C0032 (the Triton IT foothold, 2014-10 to 2017-01)	Not named in the advisories opened	[82] [85] [86] [94] [101]
T1572	Protocol Tunneling	Command and Control	Ember Bear; C0032 (the Triton IT foothold, 2014-10 to 2017-01); C0034 2022 Ukraine Electric Power Attack (2022-06 to 2022-10)	AA23-129A (2023-05-09); AA23-347A (2023-12-12); AA24-249A (2024-09-05)	[38] [42] [51] [86] [94] [101]
T1573	Encrypted Channel	Command and Control	APT29; C0030 Triton SIS Attack (2017-06 to 2017-08)	AA23-129A (2023-05-09); AA25-141A (2025-05)	[38] [55] [81] [94] [101]
T1573.001	Encrypted Channel: Symmetric Cryptography	Command and Control	APT28	AA23-129A (2023-05-09)	[38] [80] [101]
T1573.002	Encrypted Channel: Asymmetric Cryptography	Command and Control	Not recorded for these groups or campaigns	AA23-129A (2023-05-09)	[38]
T1574.001	Hijack Execution Flow: DLL	Stealth, Execution	Not recorded for these groups or campaigns	AA23-347A (2023-12-12, stated as T1574.002); AA25-141A (2025-05)	[42] [55]
T1583	Acquire Infrastructure	Resource Development	Sandworm Team; Ember Bear; Star Blizzard	Not named in the advisories opened	[82] [86] [87] [101]
T1583.001	Acquire Infrastructure: Domains	Resource Development	APT28; Sandworm Team; Dragonfly; Gamaredon Group; Star Blizzard; C0023 Operation Ghost (2013-09 to 2019-10); C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA23-341A (2023-12-07)	[40] [80] [82] [84] [85] [87] [94] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1583.003	Acquire Infrastructure: Virtual Private Server	Resource Development	APT28; Dragonfly; Gamaredon Group; Ember Bear; C0032 (the Triton IT foothold, 2014-10 to 2017-01)	AA24-249A (2024-09-05)	[51] [80] [84] [85] [86] [94] [101]
T1583.004	Acquire Infrastructure: Server	Resource Development	Sandworm Team	Not named in the advisories opened	[82] [101]
T1583.006	Acquire Infrastructure: Web Services	Resource Development	APT28; Turla; APT29; Gamaredon Group	Not named in the advisories opened	[80] [81] [83] [85] [101]
T1584	Compromise Infrastructure	Resource Development	C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	AA23-129A (2023-05-09)	[38] [94] [97]
T1584.001	Compromise Infrastructure: Domains	Resource Development	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]
T1584.003	Compromise Infrastructure: Virtual Private Server	Resource Development	Turla	Not named in the advisories opened	[83] [101]
T1584.004	Compromise Infrastructure: Server	Resource Development	Turla; Sandworm Team; Dragonfly	Not named in the advisories opened	[82] [83] [84] [101]
T1584.005	Compromise Infrastructure: Botnet	Resource Development	Sandworm Team	Not named in the advisories opened	[82] [101]
T1584.006	Compromise Infrastructure: Web Services	Resource Development	Turla	Not named in the advisories opened	[83] [101]
T1584.008	Compromise Infrastructure: Network Devices	Resource Development	APT28	Not named in the advisories opened	[80] [101]
T1585	Establish Accounts	Resource Development	Ember Bear	Not named in the advisories opened	[86] [101]
T1585.001	Establish Accounts: Social Media Accounts	Resource Development	Sandworm Team; Star Blizzard; C0023 Operation Ghost (2013-09 to 2019-10)	AA23-341A (2023-12-07)	[40] [82] [87] [94] [101]
T1585.002	Establish Accounts: Email Accounts	Resource Development	Sandworm Team; Star Blizzard	AA23-341A (2023-12-07)	[40] [82] [87] [101]
T1586.001	Compromise Accounts: Social Media Accounts	Resource Development	Sandworm Team	Not named in the advisories opened	[82] [101]
T1586.002	Compromise Accounts: Email Accounts	Resource Development	APT28; APT29; Star Blizzard	AA23-341A (2023-12-07); AA25-141A (2025-05)	[40] [55] [80] [81] [87] [101]
T1586.003	Compromise Accounts: Cloud Accounts	Resource Development	APT29	AA25-141A (2025-05)	[55] [81] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1587.001	Develop Capabilities: Malware	Resource Development	Turla; APT29; Sandworm Team; C0023 Operation Ghost (2013-09 to 2019-10); C0024 SolarWinds Compromise (2019-08 to 2021-01); C0030 Triton SIS Attack (2017-06 to 2017-08)	AA22-011A (2022-01-11); AA23-129A (2023-05-09)	[26] [38] [81] [82] [83] [94] [101]
T1587.003	Develop Capabilities: Digital Certificates	Resource Development	APT29; Gamaredon Group	Not named in the advisories opened	[81] [85] [101]
T1588	Obtain Capabilities	Resource Development	Not recorded for these groups or campaigns	AA23-129A (2023-05-09)	[38]
T1588.001	Obtain Capabilities: Malware	Resource Development	Turla; Ember Bear	AA24-249A (2024-09-05)	[51] [83] [86] [101]
T1588.002	Obtain Capabilities: Tool	Resource Development	APT28; Turla; APT29; Sandworm Team; Dragonfly; Gamaredon Group; Star Blizzard; C0030 Triton SIS Attack (2017-06 to 2017-08); C0032 (the Triton IT foothold, 2014-10 to 2017-01)	Not named in the advisories opened	[80] [81] [82] [83] [84] [85] [87] [94] [101]
T1588.005	Obtain Capabilities: Exploits	Resource Development	Ember Bear	AA24-249A (2024-09-05)	[51] [86] [101]
T1588.006	Obtain Capabilities: Vulnerabilities	Resource Development	Sandworm Team	Not named in the advisories opened	[82] [101]
T1588.007	Obtain Capabilities: Artificial Intelligence	Resource Development	APT28	Not named in the advisories opened	[80] [101]
T1589	Gather Victim Identity Information	Reconnaissance	Star Blizzard	AA23-341A (2023-12-07)	[40] [87] [101]
T1589.001	Gather Victim Identity Information: Credentials	Reconnaissance	APT28; C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[80] [94] [101]
T1589.002	Gather Victim Identity Information: Email Addresses	Reconnaissance	Sandworm Team	AA25-141A (2025-05)	[55] [82] [101]
T1589.003	Gather Victim Identity Information: Employee Names	Reconnaissance	Sandworm Team	Not named in the advisories opened	[82] [101]
T1590	Gather Victim Network Information	Reconnaissance	Not recorded for these groups or campaigns	AA23-347A (2023-12-12)	[42]
T1590.001	Gather Victim Network Information: Domain Properties	Reconnaissance	Sandworm Team	Not named in the advisories opened	[82] [101]
T1590.002	Gather Victim Network Information: DNS	Reconnaissance	Not recorded for these groups or campaigns	AA24-249A (2024-09-05)	[51]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1590.004	Gather Victim Network Information: Network Topology	Reconnaissance	Not recorded for these groups or campaigns	AA23-347A (2023-12-12)	[42]
T1591	Gather Victim Org Information	Reconnaissance	APT28	AA25-141A (2025-05)	[55] [80] [101]
T1591.002	Gather Victim Org Information: Business Relationships	Reconnaissance	Sandworm Team; Dragonfly	AA25-141A (2025-05)	[55] [82] [84] [101]
T1591.004	Gather Victim Org Information: Identify Roles	Reconnaissance	Not recorded for these groups or campaigns	AA25-141A (2025-05)	[55]
T1592	Gather Victim Host Information	Reconnaissance	Not recorded for these groups or campaigns	AA25-141A (2025-05)	[55]
T1592.002	Gather Victim Host Information: Software	Reconnaissance	Sandworm Team	AA23-347A (2023-12-12)	[42] [82] [101]
T1593	Search Open Websites/Domains	Reconnaissance	Sandworm Team; Star Blizzard	AA23-341A (2023-12-07)	[40] [82] [87] [101]
T1594	Search Victim-Owned Websites	Reconnaissance	Sandworm Team	Not named in the advisories opened	[82] [101]
T1595	Active Scanning	Reconnaissance	C0030 Triton SIS Attack (2017-06 to 2017-08)	AA24-249A (2024-09-05)	[51] [94]
T1595.001	Active Scanning: Scanning IP Blocks	Reconnaissance	Ember Bear	AA24-249A (2024-09-05)	[51] [86] [101]
T1595.002	Active Scanning: Vulnerability Scanning	Reconnaissance	APT28; APT29; Sandworm Team; Dragonfly; Ember Bear	AA22-011A (2022-01-11); AA24-249A (2024-09-05)	[26] [51] [80] [81] [82] [84] [86] [101]
T1596	Search Open Technical Databases	Reconnaissance	APT28	Not named in the advisories opened	[80] [101]
T1596.005	Search Open Technical Databases: Scan Databases	Reconnaissance	Not recorded for these groups or campaigns	AA24-249A (2024-09-05)	[51]
T1598	Phishing for Information	Reconnaissance	APT28	AA22-011A (2022-01-11)	[26] [80] [101]
T1598.002	Phishing for Information: Spearphishing Attachment	Reconnaissance	Dragonfly; Star Blizzard	Not named in the advisories opened	[84] [87] [101]
T1598.003	Phishing for Information: Spearphishing Link	Reconnaissance	APT28; Sandworm Team; Dragonfly; Star Blizzard	Not named in the advisories opened	[80] [82] [84] [87] [101]
T1606.001	Forge Web Credentials: Web Cookies	Credential Access	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1606.002	Forge Web Credentials: SAML Tokens	Credential Access	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]
T1608	Stage Capabilities	Resource Development	Not recorded for these groups or campaigns	AA23-129A (2023-05-09)	[38]
T1608.001	Stage Capabilities: Upload Malware	Resource Development	Sandworm Team; Gamaredon Group; Star Blizzard	Not named in the advisories opened	[82] [85] [87] [101]
T1608.004	Stage Capabilities: Drive-by Target	Resource Development	Dragonfly	Not named in the advisories opened	[84] [101]
T1610	Deploy Container	Execution	Not recorded for these groups or campaigns	AA23-129A (2023-05-09)	[38]
T1615	Group Policy Discovery	Discovery	Turla	Not named in the advisories opened	[83] [101]
T1620	Reflective Code Loading	Stealth	Gamaredon Group	Not named in the advisories opened	[85] [101]
T1621	Multi-Factor Authentication Request Generation	Credential Access	APT29	AA24-057A (2024-02-26)	[45] [81] [101]
T1649	Steal or Forge Authentication Certificates	Credential Access	APT29	Not named in the advisories opened	[81] [101]
T1651	Cloud Administration Command	Execution	APT29	Not named in the advisories opened	[81] [101]
T1654	Log Enumeration	Discovery	Ember Bear	AA24-249A (2024-09-05)	[51] [86] [101]
T1659	Content Injection	Initial Access, Command and Control	Not recorded for these groups or campaigns	AA25-141A (2025-05)	[55]
T1665	Hide Infrastructure	Command and Control	APT29; C0024 SolarWinds Compromise (2019-08 to 2021-01)	AA25-141A (2025-05)	[55] [81] [94] [101]
T1669	Wi-Fi Networks	Initial Access	APT28; C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	Not named in the advisories opened	[80] [94] [97] [101]
T1680	Local Storage Discovery	Discovery	C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[94]
T1684.001	Social Engineering: Impersonation	Stealth	APT28; Star Blizzard	Not named in the advisories opened	[80] [87] [101]

ID	Technique	Tactic (v20)	Recorded by ATT&CK for	Stated in advisory (date)	Sources
T1685	Disable or Modify Tools	Defense Impairment	Turla; Gamaredon Group; C0024 SolarWinds Compromise (2019-08 to 2021-01); C0028 2015 Ukraine Electric Power Attack (2015-12 to 2016-01)	AA23-347A (2023-12-12, stated as T1562.001)	[42] [83] [85] [94] [100] [101]
T1685.001	Disable or Modify Tools: Disable or Modify Windows Event Log	Defense Impairment	C0024 SolarWinds Compromise (2019-08 to 2021-01); C0025 2016 Ukraine Electric Power Attack (2016-12)	Not named in the advisories opened	[94]
T1685.002	Disable or Modify Tools: Disable or Modify Cloud Log	Defense Impairment	APT29	Not named in the advisories opened	[81] [101]
T1685.005	Disable or Modify Tools: Clear Windows Event Logs	Defense Impairment	APT28; Dragonfly	AA25-141A (2025-05, stated as T1070.001)	[55] [80] [84] [100] [101]
T1686	Disable or Modify System Firewall	Defense Impairment	Dragonfly; C0024 SolarWinds Compromise (2019-08 to 2021-01)	Not named in the advisories opened	[84] [94] [101]
T1686.003	Disable or Modify System Firewall: Windows Host Firewall	Defense Impairment	C0051 APT28 Nearest Neighbor (2022-02 to 2024-11)	Not named in the advisories opened	[94] [97]

Enterprise techniques inferred by this study

These are this study's mappings from a dated narrative to a technique. They are **Inferred**, carry the lowest score in the Navigator layer, and should not be quoted as ATT&CK or advisory mappings.

ID	Technique	Tactic (v20)	Narrative it rests on (source, date)	Sources
T1602.002	Data from Configuration Repository: Network Device Configuration Dump	Collection	FSB Center 16 collected configuration files from thousands of US networking devices (FBI PSA, 2025-08-20) and copied router configurations worldwide (AA26-194A, 2026-07-13)	[56] [62]
T1557	Adversary-in-the-Middle	Credential Access, Collection	Secret Blizzard used an ISP-level adversary-in-the-middle position against embassies in Moscow (Microsoft, 2025-07-31); APT28 hijacked DHCP and DNS on SOHO routers for adversary-in-the-middle attacks on Outlook Web Access (FBI PSA, 2026-04-07)	[77] [59]

ATT&CK for ICS

ATT&CK records ICS techniques for three of the groups at group level (**Confirmed** [95] [100]). The ICS techniques of the Ukraine grid campaigns (C0028, C0025, C0034) are bound in SANDWORM_OPERATIONS.md, and the Triton campaign's are on the generated campaign pages C0030 and C0032. The library has no generated ICS technique pages, so IDs link to attack.mitre.org; the ICS ATT&CK Technique Atlas groups them by tactic.

ID	Technique	Tactic	Recorded for	Sources
T0819	Exploit Public-Facing Application	Initial Access	Sandworm Team	[82] [100]
T0884	Connection Proxy	Command and Control	Sandworm Team	[82] [100]
T1701.001	Command and Scripting Interpreter: PowerShell	Execution	Sandworm Team	[82] [100]
T1701.002	Command and Scripting Interpreter: Windows Command Shell	Execution	Sandworm Team	[82] [100]
T0817	Drive-by Compromise	Initial Access	Dragonfly; TEMP.Veles	[84] [100]
T0862	Supply Chain Compromise	Initial Access	Dragonfly; TEMP.Veles	[84] [100]
T0840	Network Connection Enumeration	Discovery	Stated in the AA23-129A table for Snake (2023-05-09); no group record	[38] [100]

ALLANITE's four ICS techniques (T0817, T0852, T0859, T0865) are in the library's ICS data, but the group has no generated page and no government attribution was opened for it, so it is listed here only for completeness (**Reported** [95]).

Mobile

ID	Technique	Tactic	Recorded for or stated by	Sources
T1409	Stored Application Data	Collection	Sandworm Team	[82] [100]
T1660	Phishing	Initial Access	Sandworm Team	[82] [100]
T1676	Linked Devices	Collection, Persistence	Sandworm Team; Star Blizzard	[82] [87] [100]
T1627	Execution Guardrails	Defense Evasion (Mobile)	Stated in AA25-141A (2025-05)	[55] [100]
T1627.001	Execution Guardrails: Geofencing	Defense Evasion (Mobile)	Stated in AA25-141A (2025-05)	[55] [100]

The Mobile domain still uses the Defense Evasion tactic name in v20.0 (**Confirmed** [100]).

Groups, campaigns and software

Object	ID	Relationship in ATT&CK v20.0 and the record	Sources
APT28	G0007	GRU Unit 26165; 93 techniques; campaign C0051	[80]
Sandworm Team	G0034	GRU Unit 74455; 79 Enterprise, 4 ICS, 3 Mobile techniques; campaigns C0028, C0025, C0034	[82]
Ember Bear	G1003	GRU Unit 29155; 47 techniques	[86]
APT29	G0016	SVR; 66 techniques; campaigns C0023, C0024	[81]
Turla	G0010	FSB; 68 techniques; no campaigns	[83]
Dragonfly	G0035	FSB Center 16; 56 Enterprise, 2 ICS techniques	[84]
Star Blizzard	G1033	Russia (governments: FSB Centre 18); 20 Enterprise, 1 Mobile technique	[87] [40]
Gamaredon Group	G0047	Suspected Russian (Ukraine: FSB Center 18); 70 techniques	[85]
TEMP.Veles	G0088	Russia-based (AA22-110A: TsNIIKhM); 2 ICS techniques; campaigns C0030, C0032	[94] [30]
Saint Bear	G1031	Russian-nexus; distinct from Ember Bear	[88]
Winter Vivern	G1035	No explicit state attribution	[89]
Wizard Spider	G0102	Russia-based criminal	[93]
Indrik Spider	G0119	Russia-based criminal (Evil Corp)	[93]
ALLANITE	G1000	ATT&CK for ICS only; no library page	[95]
Operation Ghost	C0023	APT29, 2013-09 to 2019-10	[94]
SolarWinds Compromise	C0024	APT29, 2019-08 to 2021-01; home study SOLARWINDS_SUNBURST.md	[94]
2015 Ukraine Electric Power Attack	C0028	Sandworm Team, 2015-12 to 2016-01	[94]
2016 Ukraine Electric Power Attack	C0025	Sandworm Team, 2016-12	[94]
2022 Ukraine Electric Power Attack	C0034	Sandworm Team, 2022-06 to 2022-10	[94]
APT28 Nearest Neighbor Campaign	C0051	APT28, 2022-02 to 2024-11	[97]
C0032	C0032	TEMP.Veles, 2014-10 to 2017-01; the Triton IT foothold before the safety-system attack	[94]
Triton Safety Instrumented System Attack	C0030	TEMP.Veles, 2017-06 to 2017-08	[94]
2025 Poland Wiper Attacks	C0063	No group associated; ATT&CK cites FSB Dragonfly and GRU Sandworm assessments; 53 techniques in the library data	[90] [98] [94]

Software	ID	Used by (library data and the record)	Sources
CHOPSTICK	S0023	APT28	[93]
Zebrocy	S0251	APT28	[93]
LoJax	S0397	APT28	[93]
Drovorub	S0502	APT28	[93]
LAMEHUG	S9035	APT28	[93] [80]
BlackEnergy	S0089	Sandworm Team	[93]
Industroyer, Industroyer2	S0604, S1072	Sandworm Team	[93]
NotPetya	S0368	Sandworm Team	[93]
Olympic Destroyer	S0365	Sandworm Team	[93]
VPNFilter, Cyclops Blink	S1010, S0687	Sandworm Team	[93]
AcidRain, AcidPour	S1125, S1167	Sandworm Team	[93]
Prestige, Kapeka, CaddyWiper	S1058, S1190, S0693	Sandworm Team (CaddyWiper on the live page only)	[93] [82]
WhisperGate	S0689	Ember Bear; not unique to Unit 29155 per AA24-249A	[86] [51]
SUNBURST, SUNSPOT, TEARDROP	S0559, S0562, S0560	APT29 (SolarWinds)	[93]
GoldMax, FoggyWeb	S0588, S0661	APT29	[93]
WellMess, WellMail	S0514, S0515	APT29 (vaccine campaign, 2020)	[12] [93]
Uroburos, Carbon, Kazuar, ComRAT, LightNeuron, TinyTurla	S0022, S0335, S0265, S0126, S0395, S0668	Turla	[93]
Spica	S1140	Star Blizzard	[87]
Saint Bot, OutSteel	S1018, S1017	Saint Bear	[88]
DynoWiper, LazyWiper	S9038, S9039	Campaign C0063	[98]
Mimikatz, PsExec	S0002, S0029	Six of the groups each	[93]
Net, Impacket, reGeorg, Reg	S0039, S0357, S1187, S0075	Five, four, three and three groups	[93]
TrickBot, Conti, Ryuk	S0266, S0575, S0446	Wizard Spider	[93]
Dridex, BitPaymer, WastedLocker	S0384, S0570, S0612	Indrik Spider	[93]
Emotet, Black Basta	S0367, S1070	Criminal ecosystem named in AA22-110A and AA24-131A	[30] [49]

The shared layer of tooling is commodity and living-off-the-land software (Mimikatz, PsExec, Net, Impacket); the bespoke implants are group-specific (**Inferred** from [93]). Triton (S1009) is an ICS-domain software object without a library page (**Confirmed** [100]).

CWE and CAPEC

CWE IDs come from the `cwes` field of the KEV feed for each exploited CVE (Part 5). CAPEC IDs come from the library's own data: `data/weaknesses/cwe.jsonl` for weakness-to-pattern links, and the `capec` field of `data/attack/technique_profiles.jsonl` for technique-to-pattern links (**Confirmed** [92] [105] [100]). The broad classes CWE-20, CWE-74, CWE-119 and CWE-284 are not expanded, because their pattern lists say nothing specific about these flaws. See the CWE Reference and CAPEC Reference.

Weakness	CVEs in this record	CAPEC patterns linked in the library data
CWE-22 Path Traversal	CVE-2018-13379, CVE-2019-11510, CVE-2019-19781, CVE-2020-5902	CAPEC-126 Path Traversal; CAPEC-64; CAPEC-76; CAPEC-78; CAPEC-79
CWE-78 OS Command Injection	CVE-2019-10149, CVE-2020-4006, CVE-2020-12641	CAPEC-88 OS Command Injection; CAPEC-6; CAPEC-15; CAPEC-43; CAPEC-108
CWE-79 Cross-site Scripting	CVE-2020-35730	CAPEC-63; CAPEC-85; CAPEC-209; CAPEC-588; CAPEC-591; CAPEC-592
CWE-89 SQL Injection	CVE-2021-44026	CAPEC-66 SQL Injection; CAPEC-7; CAPEC-108; CAPEC-109; CAPEC-110; CAPEC-470
CWE-94 Code Injection	CVE-2019-7609, CVE-2022-3236	CAPEC-242 Code Injection; CAPEC-35; CAPEC-77
CWE-611 XML External Entity	CVE-2019-9670	CAPEC-221
CWE-287 Improper Authentication	CVE-2020-0688, CVE-2021-33044, CVE-2021-33045	CAPEC-114 Authentication Abuse; CAPEC-115 Authentication Bypass; CAPEC-151; CAPEC-194; CAPEC-22; CAPEC-57; CAPEC-593; CAPEC-633; CAPEC-650; CAPEC-94
CWE-288 Authentication Bypass Using an Alternate Path	CVE-2023-42793	CAPEC-127; CAPEC-665
CWE-294 Authentication Bypass by Capture-replay	CVE-2023-23397	CAPEC-102; CAPEC-509; CAPEC-555; CAPEC-561; CAPEC-60; CAPEC-644 Pass the Hash; CAPEC-645; CAPEC-652; CAPEC-701; CAPEC-94
CWE-290 Authentication Bypass by Spoofing	CVE-2023-50224	CAPEC-21; CAPEC-22; CAPEC-459; CAPEC-461; CAPEC-473; CAPEC-476; CAPEC-59; CAPEC-60; CAPEC-667; CAPEC-94
CWE-798 Hard-coded Credentials	CVE-2022-26138	CAPEC-191; CAPEC-70
CWE-918 Server-Side Request Forgery	CVE-2021-26855	CAPEC-664
CWE-330 Insufficiently Random Values	CVE-2020-1472	CAPEC-112 Brute Force; CAPEC-485; CAPEC-59
CWE-352 Cross-Site Request Forgery	CVE-2008-4128	CAPEC-62 Cross Site Request Forgery; CAPEC-111; CAPEC-462; CAPEC-467

Weakness	CVEs in this record	CAPEC patterns linked in the library data
CWE-917 Expression Language Injection; CWE-351 Insufficient Type Distinction	CVE-2022-26134; CVE-2023-38831	None in the library data

Technique (use in this record)	CAPEC patterns linked in the library data
`T1110` Brute Force; `T1110.003` Password Spraying (SVR, GRU)	CAPEC-112 Brute Force; CAPEC-565 Password Spraying
`T1078` Valid Accounts	CAPEC-560 Use of Known Domain Credentials
`T1133` External Remote Services; `T1114.002` Remote Email Collection	CAPEC-555 Remote Services with Stolen Credentials
`T1566.001`, `T1566.002` Spearphishing (APT28, Star Blizzard)	CAPEC-163 Spear Phishing
`T1539` Steal Web Session Cookie; `T1528` Steal Application Access Token	CAPEC-21; CAPEC-31 (cookie only)
`T1550.004` Web Session Cookie (Star Blizzard)	CAPEC-60 Reusing Session IDs
`T1550.002` Pass the Hash (Unit 29155)	CAPEC-644 Use of Captured Hashes
`T1505.003` Web Shell	CAPEC-650 Upload a Web Shell to a Web Server
`T1014` Rootkit (Snake)	CAPEC-552 Install Rootkit
`T1195.002` Compromise Software Supply Chain (SolarWinds)	CAPEC-186 Malicious Software Update; CAPEC-442; CAPEC-443; CAPEC-523; CAPEC-669; CAPEC-673; CAPEC-678; CAPEC-691
`T1557` Adversary-in-the-Middle (Inferred)	CAPEC-94 Adversary in the Middle
`T1602` Data from Configuration Repository, parent of the inferred `T1602.002`	CAPEC-150 Collect Data from Common Resource Locations

D3FEND countermeasures

D3FEND countermeasures below come from the library's `data/attack/technique_to_d3fend.jsonl`, and each links to its library page; the table lists the countermeasures that match the advisories' own recommendations, not every row the library holds (**Confirmed** [102]). See the D3FEND Reference.

Behavior in the record	Techniques	D3FEND countermeasures in the library data
Password spraying and credential theft (SVR, GRU)	`T1110.003`	Multi-factor Authentication (`D3-MFA`), Credential Hardening (`D3-CH`), Strong Password Policy (`D3-SPP`), Connection Attempt Analysis (`D3-CAA`), Decoy User Credential (`D3-DUC`), User Geolocation Logon Pattern Analysis (`D3-UGLPA`)
Valid and dormant accounts, cloud accounts	`T1078`, `T1078.004`	Account Locking (`D3-AL`), User Account Permissions (`D3-UAP`)

Behavior in the record	Techniques	D3FEND countermeasures in the library data
Session-cookie and token theft (Star Blizzard, SVR)	T1539, T1550.004, T1528	Authentication Cache Invalidation (D3-ANCI), Credential Revocation (D3-CR), Credential Rotation (D3-CRO), Token Binding (D3-TB, T1528 only)
Mail forwarding rules (Star Blizzard)	T1114.003	Application Configuration Hardening (D3-ACH), Configuration Inventory (D3-CI), Disable Remote Access (D3-DRA)
Exploiting internet-facing applications	T1190	Inbound Traffic Filtering (D3-ITF), Network Traffic Filtering (D3-NTF), Network Traffic Signature Analysis (D3-NTSA), Application Protocol Command Analysis (D3-APCA)
Spearphishing	T1566.001, T1566.002	Email Filtering (D3-EF), Dynamic Analysis (D3-DA), Content Quarantine (D3-CQ), Homoglyph Detection (D3-HD), URL Analysis (D3-UA, links only)
Forced NTLM authentication (CVE-2023-23397)	T1187	File Integrity Monitoring (D3-FIM); the advisory's own control, blocking outbound SMB and NTLM, matches Outbound Traffic Filtering (D3-OTF), which the library does not map to T1187 (**Inferred**)
Proxy networks and tunnels (EdgeRouters, SOHO routers, Snake)	T1090.003, T1572	Outbound Traffic Filtering (D3-OTF), Relay Pattern Analysis (D3-RPA), Protocol Metadata Anomaly Detection (D3-PMAD)
Web shells (Unit 29155)	T1505.003	File Integrity Monitoring (D3-FIM), Executable Allowlisting (D3-EAL), Process Spawn Analysis (D3-PSA), Process Lineage Analysis (D3-PLA)
LSASS credential dumping	T1003.001	Process Spawn Analysis (D3-PSA), System Call Filtering (D3-SCF)
Kernel rootkit (Snake)	T1014	Firmware Verification (D3-FV), Software Inventory (D3-SWI), Software Update (D3-SU)
Software supply chain	T1195.002	Software Update (D3-SU), Software Inventory (D3-SWI), Asset Vulnerability Enumeration (D3-AVE)
Disk wipers (WhisperGate, NotPetya)	T1561.002	Disk Partitioning (D3-DKP)
External remote services	T1133	Session Termination (D3-ST), the only row in the library data
No countermeasure in the library data	T1059.001, T1588.002, T1595.002, T1098.005, T1110, T1485, T1550.002, T1606.002, T1685.005	See Part 7

Some library rows pair a technique with countermeasures that do not fit it closely (for example, T1621 MFA Request Generation maps to process-isolation countermeasures), so the table above picks the rows that match an advisory's recommendation (**Inferred** from [102]).

Navigator layer

The study's techniques are collected in an ATT&CK Navigator layer, `navigator/library-layers/threat-actors-russia.json`, listed with the library's other curated layers on the Navigator layers page. Open it in the ATT&CK Navigator. It holds the 330 techniques of the Enterprise table and the two inferred techniques, each shown under all its v20 tactics. Scores mean evidence basis, not frequency or severity: 100 for techniques named in a joint government advisory, 60 for techniques ATT&CK records for one of the nine groups or eight campaigns but no verified advisory names, and 30 for this study's inferences. Each technique's comment names the study's source numbers. Campaign C0063 is left out because ATT&CK associates it with no group.

Cross-cutting themes

- **Attribution is a ladder, and this record reaches its top rungs.** Governments name units, grand juries name officers and sanctions name organisations. Even so, the rungs differ: a grand jury's allegation, a government's "almost certainly", a vendor's "medium confidence", ATT&CK's "suspected". Carry the rung with the name. See the Threat Intelligence Reference.
- **Services compete and overlap.** Three GRU units, two FSB centers and the SVR run parallel programs, sometimes against the same targets, and the Poland 2025 attack shows experienced analysts assigning one event to two services. Merging indicators across those names without checking the source's boundary produces false confidence (**Inferred** from [79] [58] [61] [51]).
- **The edge is the battlefield.** Routers, VPNs, mail servers, cameras and CI servers appear in nearly every advisory from 2018 to 2026, and the network-device campaigns of the GRU and the FSB run in parallel (**Inferred** from Part 5).
- **Identity is the SVR's main door.** The SVR's recent record at Microsoft, HPE and in AA24-057A is password spraying, dormant accounts, tokens, OAuth applications and device registration, alongside opportunistic exploitation such as TeamCity (**Confirmed** [68] [43] [45] [42]). See the Identity Security Reference.
- **Fronts and criminals blur the line.** Unit 74455 likely supported a hacktivist front; Treasury links Trickbot members to the intelligence services; KillNet's ties are unconfirmed. For a hospital, the distinction between a state unit and a state-tolerated crew matters less than the controls that stop both (**Confirmed** [57] [34] [33]; the conclusion is **Inferred**). See the Ransomware Defense Reference.
- **Legal action has not stopped the tempo.** Six rounds of US indictments since 2018, multiple sanctions packages, two domain seizures and a router-network takedown sit in the record alongside continuing operations into 2026 (**Inferred** from the timeline).
- **The library's own data has edges worth knowing.** The library lacks a generated page for ALLANITE and for the ICS software Triton; its copy of the APT28 and Sandworm software lists differs slightly from the live pages; and its D3FEND mapping leaves several common techniques without a countermeasure. Readers comparing tools against this layer should know which counts they are using (**Confirmed** [80] [82] [96] [102]).

What is overstated and what is understated

- **Overstated: Russian state units as the attackers of hospitals.** None of the government documents opened for this study attributes a disruptive attack on a hospital to a Russian state unit. The state record on healthcare is espionage, collateral damage and sector lists; the attacks that disrupted care came from criminal ransomware crews and pro-Russia hacktivists (**Inferred** from [12] [32] [51] [62] [33] [34]; see "What the record does not show" in Part 6).
- **Overstated: vendor names as units.** AA24-249A warns that vendor names may not map one-to-one to its own attribution, and the FBI calls Berserk Bear and Dragonfly separate but related clusters (**Confirmed** [51] [56]). A vendor name marks a cluster of activity, not a unit (**Inferred**).
- **Overstated: KillNet's own claims.** HC3 says KillNet's ties to the FSB or SVR are unconfirmed and advises treating its claims "with a grain of salt" (**Confirmed** [33]). The January 2023 DDoS wave on hospitals is in the record; a claimed compromise is a lead to check, not an incident (**Inferred** from [33] [35]).
- **Understated: patch debt on old KEV entries.** CVE-2018-0171 has been in KEV since 3 November 2021, and the FBI reported FSB Center 16 exploiting it in August 2025 (**Confirmed** [92] [56]). Attention goes to new flaws, while this record runs on old, known ones (**Inferred**).
- **Understated: end-of-life network devices.** TA18-106A warned in 2018 about Smart Install, SNMP and GRE on routers; the FBI's 2025 notice reports Smart Install and SNMP exploitation on end-of-life devices, and AA26-194A asked for router hygiene again in 2026 (**Confirmed** [6] [56] [62]). Devices that no longer receive firmware fall outside a CVE-driven patch program (**Inferred**).
- **Understated: identity conditions over CVEs.** The SVR's access paths in AA24-057A are password spraying, dormant accounts, tokens and device registration, and Midnight Blizzard entered Microsoft through a legacy test tenant account (**Confirmed** [45] [69]). None of those conditions has a CVE, so a CVE-only backlog does not show them (**Inferred**).

Sources

Every source below was opened on 3 November 2026, except rows 106 and 107, which were read on 10 November 2026. Grades apply to the source type: Confirmed for a primary (the publisher of the fact), Reported for a secondary account or a summary of a document that was not itself opened.

#	Source	URL	Opened	Grade
1	WADA, WADA Confirms Attack by Russian Cyber Espionage Group (2016-09-13)	https://www.wada-ama.org/en/news/wada-confirms-attack-russian-cyber-espionage-group	2026-11-03	Confirmed
2	DHS and FBI, Joint Analysis Report JAR-16-20296A, GRIZZLY STEPPE: Russian Malicious Cyber Activity (2016-12-29)	https://www.cisa.gov/sites/default/files/publications/JAR_16-20296A_GRIZZLY%20STEPPE-2016-1229.pdf	2026-11-03	Confirmed
3	The White House (archived), Statement from the Press Secretary on NotPetya (2018-02-15)	https://trumpwhitehouse.archives.gov/briefings-statements/statement-press-secretary-25/	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
4	GOV.UK, Foreign Office Minister condemns Russia for NotPetya attacks (2018-02-15)	https://www.gov.uk/government/news/foreign-office-minister-condemns-russia-for-notpetya-attacks	2026-11-03	Confirmed
5	DHS and FBI, Alert TA18-074A, Russian Government Cyber Activity Targeting Energy and Other Critical Infrastructure Sectors (2018-03-15)	https://www.cisa.gov/news-events/alerts/2018/03/15/russian-government-cyber-activity-targeting-energy-and-other-critical-infrastructure-sectors	2026-11-03	Confirmed
6	DHS, FBI and NCSC, Alert TA18-106A, Russian State-Sponsored Cyber Actors Targeting Network Infrastructure Devices (2018-04-16, updated 2018-04-19)	https://www.cisa.gov/news-events/alerts/2018/04/16/russian-state-sponsored-cyber-actors-targeting-network-infrastructure-devices	2026-11-03	Confirmed
7	FBI, Russian Interference in 2016 U.S. Elections (wanted notice for the 2018-07-13 indictment)	https://www.fbi.gov/wanted/cyber/russian-interference-in-2016-u-s-elections	2026-11-03	Confirmed
8	FBI, wanted notice for Aleksey Viktorovich Lukashev	https://www.fbi.gov/wanted/cyber/aleksey-viktorovich-lukashev	2026-11-03	Confirmed
9	FBI, GRU Hacking to Undermine Anti-Doping Efforts (wanted notice for the 2018-10-03 indictment)	https://www.fbi.gov/wanted/cyber/gru-hacking-to-undermine-anti-doping-efforts	2026-11-03	Confirmed
10	US Department of State (archived), The United States Condemns Russian Cyber Attack Against the Country of Georgia (2020-02-20)	https://2017-2021.state.gov/the-united-states-condemns-russian-cyber-attack-against-the-country-of-georgia/index.html	2026-11-03	Confirmed
11	GOV.UK, UK condemns Russia's GRU over Georgia cyber-attacks (2020-02-20)	https://www.gov.uk/government/news/uk-condemns-russias-gru-over-georgia-cyber-attacks	2026-11-03	Confirmed
12	NCSC and CSE, with NSA and CISA, Advisory: APT29 targets COVID-19 vaccine development, v1.1 (2020-07-16), PDF; also served at https://www.ncsc.gov.uk/files/Advisory-APT29-targets-COVID-19-vaccine-development-V1-1.pdf	https://www.ncsc.gov.uk/sites/default/files/documents/Advisory-APT29-targets-COVID-19-vaccine-development-V1-1.pdf	2026-11-03	Confirmed
13	NCSC, Advisory: APT29 targets COVID-19 vaccine development (news page, 2020-07-16)	https://www.ncsc.gov.uk/news/advisory-apt29-targets-covid-19-vaccine-development	2026-11-03	Confirmed
14	FBI, GRU Hackers' Destructive Malware and International Cyber Attacks (wanted notice for the 2020-10-15 indictment)	https://www.fbi.gov/wanted/cyber/gru-hackers-destructive-malware-and-international-cyber-attacks	2026-11-03	Confirmed
15	GOV.UK, UK exposes series of Russian cyber attacks against Olympic and Paralympic Games (2020-10-19)	https://www.gov.uk/government/news/uk-exposes-series-of-russian-cyber-attacks-against-olympic-and-paralympic-games	2026-11-03	Confirmed
16	FBI and CISA, AA20-296A, Russian State-Sponsored Advanced Persistent Threat Actor Compromises U.S. Government Targets (2020-10-22)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa20-296a	2026-11-03	Confirmed
17	US Treasury, press release sm1162, designation of TsNIIKhM (2020-10-23)	https://home.treasury.gov/news/press-releases/sm1162	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
18	CISA, FBI and HHS, AA20-302A, Ransomware Activity Targeting the Healthcare and Public Health Sector (2020-10-28)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa20-302a	2026-11-03	Confirmed
19	The White House (archived), Fact Sheet: Imposing Costs for Harmful Foreign Activities by the Russian Government (2021-04-15)	https://bidenwhitehouse.archives.gov/briefing-room/statements-releases/2021/04/15/fact-sheet-imposing-costs-for-harmful-foreign-activities-by-the-russian-government/	2026-11-03	Confirmed
20	US Treasury, press release jy0127, sanctions on Russian technology companies (2021-04-15)	https://home.treasury.gov/news/press-releases/jy0127	2026-11-03	Confirmed
21	GOV.UK, Russia: UK exposes Russian involvement in SolarWinds cyber compromise (2021-04-15)	https://www.gov.uk/government/news/russia-uk-exposes-russian-involvement-in-solarwinds-cyber-compromise	2026-11-03	Confirmed
22	FBI, DHS and CISA, AA21-116A, Russian SVR Targets U.S. and Allied Networks (2021-04-26)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa21-116a	2026-11-03	Confirmed
23	FBI FLASH CP-000147-MW, Conti Ransomware Attacks Impact Healthcare and First Responder Networks (2021-05-20)	https://www.ic3.gov/Media/News/2021/210521.pdf	2026-11-03	Confirmed
24	CISA bulletin announcing the NSA, CISA, FBI and NCSC advisory on a GRU 85th GTsSS brute-force campaign (2021-07-01)	https://content.govdelivery.com/accounts/USDHSCISA/bulletins/2e66dc2	2026-11-03	Confirmed
25	HSE and PwC, Conti cyber attack on the HSE: Independent Post Incident Review (2021-12-03)	https://about.hse.ie/api/v2/download-file/file_based_publications/conti_cyber_attack_on_the_hse_full_report.pdf	2026-11-03	Confirmed
26	CISA, FBI and NSA, AA22-011A, Understanding and Mitigating Russian State-Sponsored Cyber Threats to U.S. Critical Infrastructure (2022-01-11)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa22-011a	2026-11-03	Confirmed
27	GOV.UK, UK assess Russian involvement in cyber attacks on Ukraine (2022-02-18)	https://www.gov.uk/government/news/uk-assess-russian-involvement-in-cyber-attacks-on-ukraine	2026-11-03	Confirmed
28	GOV.UK, UK exposes Russian spy agency behind cyber incidents (2022-03-24)	https://www.gov.uk/government/news/uk-exposes-russian-spy-agency-behind-cyber-incidents	2026-11-03	Confirmed
29	FBI, wanted notice for Evgeny Viktorovich Gladkikh (case announced 2022-03-24)	https://www.fbi.gov/wanted/cyber/evgeny-viktorovich-gladkikh	2026-11-03	Confirmed
30	CISA and Five Eyes partners, AA22-110A, Russian State-Sponsored and Criminal Cyber Threats to Critical Infrastructure (2022-04-20, revised 2022-05-09)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa22-110a	2026-11-03	Confirmed
31	GOV.UK, Russia behind cyber-attack with Europe-wide impact an hour before Ukraine invasion (2022-05-10)	https://www.gov.uk/government/news/russia-behind-cyber-attack-with-europe-wide-impact-an-hour-before-ukraine-invasion	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
32	HHS HC3, Major Cyber Organizations of the Russian Intelligence Services (2022-05-19), AHA copy	https://www.aha.org/system/files/media/file/2022/06/Major-Cyber-Organizations-of-Russian-Intelligence-Services.pdf	2026-11-03	Confirmed
33	HHS HC3, Analyst Note: Pro-Russian Hacktivist Group KillNet Threat to HPH Sector (2023-01-30), AHA copy	https://www.aha.org/system/files/media/file/2023/01/hc3-tlp-clear-analyst-note-pro-russian-hacktivist-group-killnet-threat-to-hph-sector-1-30-23.pdf	2026-11-03	Confirmed
34	US Treasury, press release jy1256, Trickbot sanctions (2023-02-09)	https://home.treasury.gov/news/press-releases/jy1256	2026-11-03	Confirmed
35	HHS HC3, Analyst Note: KillNet's Targeting of the HPH Sector, December 2022 to March 2023 (2023-04-05), AHA copy	https://www.aha.org/system/files/media/file/2023/04/hc3-tlp-clear-analyst-note-killnets-targeting-of-the-hph-sector-december-2022-march-2023.pdf	2026-11-03	Confirmed
36	CISA, APT28 Exploits Known Vulnerability to Carry Out Reconnaissance and Deploy Malware on Cisco Routers (2023-04-18)	https://www.cisa.gov/news-events/alerts/2023/04/18/apt28-exploits-known-vulnerability-carry-out-reconnaissance-and-deploy-malware-cisco-routers	2026-11-03	Confirmed
37	NCSC, APT28 exploits known vulnerability to carry out reconnaissance and deploy malware on Cisco routers (2023-04-18)	https://www.ncsc.gov.uk/news/apt28-exploits-known-vulnerability-to-carry-out-reconnaissance-and-deploy-malware-on-cisco-routers	2026-11-03	Confirmed
38	CISA and partners, AA23-129A, Hunting Russian Intelligence "Snake" Malware (2023-05-09)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa23-129a	2026-11-03	Confirmed
39	FBI, wanted notice for Ruslan Aleksandrovich Peretyatko (indictment 2023-12-05)	https://www.fbi.gov/wanted/cyber/ruslan-aleksandrovich-peretyatko	2026-11-03	Confirmed
40	NCSC, CISA and partners, AA23-341A, Russian FSB Cyber Actor Star Blizzard Continues Worldwide Spear-phishing Campaigns (2023-12-07)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa23-341a	2026-11-03	Confirmed
41	GOV.UK, UK exposes attempted Russian cyber interference in politics and democratic processes (2023-12-07)	https://www.gov.uk/government/news/uk-exposes-attempted-russian-cyber-interference-in-politics-and-democratic-processes	2026-11-03	Confirmed
42	FBI, CISA, NSA, SKW, CERT Polska and NCSC, AA23-347A, Russian Foreign Intelligence Service (SVR) Exploiting JetBrains TeamCity CVE Globally (2023-12-12, revised 2023-12-13)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa23-347a	2026-11-03	Confirmed
43	Hewlett Packard Enterprise, Form 8-K (2024-01-19), SEC EDGAR	`https://www.sec.gov/Archives/edgar/data/1645590/000164559024000009/hpe-20240119.htm`	2026-11-03	Confirmed
44	US Department of Justice, court-authorized disruption of a botnet controlled by the Russian GRU (2024-02-15), read from an archived copy of this URL	https://www.justice.gov/archives/opa/pr/justice-department-conducts-court-authorized-disruption-botnet-controlled-russian	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
45	NCSC and partners, AA24-057A, SVR Cyber Actors Adapt Tactics for Initial Cloud Access (2024-02-26)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa24-057a	2026-11-03	Confirmed
46	FBI, NSA, US Cyber Command and partners, Russian Cyber Actors Use Compromised Routers to Facilitate Cyber Operations (2024-02-27)	https://www.ic3.gov/CSA/2024/240227.pdf	2026-11-03	Confirmed
47	Ministry of Foreign Affairs of the Czech Republic, Statement on the cyberattacks carried out by Russian actor APT28 (2024-05-03)	https://mzv.gov.cz/jnp/en/issues_and_press/press_releases/statement_of_the_mfa_on_the_cyberattacks.html	2026-11-03	Confirmed
48	US Treasury, press release jy2326, designation of LockBit leader Dmitry Khoroshev (2024-05-07)	https://home.treasury.gov/news/press-releases/jy2326	2026-11-03	Confirmed
49	CISA, FBI, HHS and MS-ISAC, AA24-131A, #StopRansomware: Black Basta (2024-05-10)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa24-131a	2026-11-03	Confirmed
50	HHS HC3, Threat Actor Profile: Midnight Blizzard (2024-06-06), AHA copy	https://www.aha.org/system/files/media/file/2024/10/hc3-threat-actor-profile-tlp-clear-midnight-blizzard-6-6-2024.pdf	2026-11-03	Confirmed
51	FBI, CISA, NSA and partners, AA24-249A, Russian Military Cyber Actors Target US and Global Critical Infrastructure (2024-09-05)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa24-249a	2026-11-03	Confirmed
52	FBI, GRU Unit 29155 Cyber Actors (wanted notice, 2024-09-05)	https://www.fbi.gov/wanted/cyber/gru-29155-cyber-actors	2026-11-03	Confirmed
53	France, Ministry for Europe and Foreign Affairs, attribution of cyberattacks against France to the Russian military intelligence service (2025-04-29)	https://www.diplomatie.gouv.fr/en/press-et-ressources/decouvrir-et-informer/actualites/russie-attribution-de-cyberattaques-contre-la-france-au-service-de-renseignement-militaire-russe	2026-11-03	Confirmed
54	ANSSI (CERT-FR), CERTFR-2025-CTI-007, targeting and compromise of French entities by the APT28 intrusion set (2025-04-29); landing page also at https://cert.ssi.gouv.fr/cti/CERTFR-2025-CTI-007/	https://cert.cyber.gouv.fr/cti/CERTFR-2025-CTI-007/	2026-11-03	Confirmed
55	NSA, FBI, CISA and partner agencies, 11 countries in all, AA25-141A, Russian GRU Targeting Western Logistics Entities and Technology Companies (May 2025, corrected 2026-04-17)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa25-141a	2026-11-03	Confirmed
56	FBI, PSA I-082025-PSA, Russian Government Cyber Actors Targeting Networking Devices, Critical Infrastructure (2025-08-20)	https://www.ic3.gov/PSA/2025/PSA250820	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
57	CISA, FBI, NSA and partners, AA25-343A, on pro-Russia hacktivists CARR, NoName057(16), Z-Pentest and Sector16 (2025-12-09)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa25-343a	2026-11-03	Confirmed
58	CERT Polska, Incident report: energy sector 2025 (2026-01-30)	https://cert.pl/en/posts/2026/01/incident-report-energy-sector-2025/	2026-11-03	Confirmed
59	FBI, PSA I-040726-PSA, on Russian GRU exploitation of SOHO routers (2026-04-07)	https://www.ic3.gov/PSA/2026/PSA260407	2026-11-03	Confirmed
60	GOV.UK, Profile: GRU cyber and hybrid threat operations (updated 2026-07-13)	https://www.gov.uk/government/publications/profile-gru-cyber-and-hybrid-threat-operations/profile-gru-cyber-and-hybrid-threat-operations	2026-11-03	Confirmed
61	GOV.UK, UK and EU strike Russian cyber networks with new sanctions (2026-07-13)	https://www.gov.uk/government/news/uk-and-eu-strike-russian-cyber-networks-with-new-sanctions	2026-11-03	Confirmed
62	CISA and 18 partner agencies, AA26-194A, Improve Router Hygiene to Protect Against Russian State-Sponsored Targeting (2026-07-13)	https://www.cisa.gov/news-events/cybersecurity-advisories/aa26-194a	2026-11-03	Confirmed
63	NCSC, UK and allies urge critical sectors to improve defences against Russian intelligence targeting (2026-07-13)	https://www.ncsc.gov.uk/news/uk-and-allies-urge-critical-sectors-to-improve-defences-against-russian-intelligence-targeting	2026-11-03	Confirmed
64	Kaspersky Securelist, The Epic Turla operation (2014-08-07)	https://securelist.com/the-epic-turla-operation/65545/	2026-11-03	Confirmed
65	FireEye (Google Cloud archive), APT28: A Window into Russia's Cyber Espionage Operations? (2014-10-27)	https://cloud.google.com/blog/topics/threat-intelligence/apt28-a-window-into-russias-cyber-espionage-operations	2026-11-03	Confirmed
66	CrowdStrike, Bears in the Midst: Intrusion into the Democratic National Committee (2016-06-14)	https://www.crowdstrike.com/en-us/blog/bears-midst-intrusion-democratic-national-committee/	2026-11-03	Confirmed
67	Microsoft, Cyberattacks targeting health care must stop (November 2020)	https://www.microsoft.com/security/blog/2020/11/18/cyberattacks-targeting-health-care-must-stop	2026-11-03	Confirmed
68	Microsoft Security Response Center, Microsoft Actions Following Attack by Nation State Actor Midnight Blizzard (2024-01-19)	https://www.microsoft.com/en-us/msrc/blog/2024/01/microsoft-actions-following-attack-by-nation-state-actor-midnight-blizzard	2026-11-03	Confirmed
69	Microsoft Threat Intelligence, Midnight Blizzard: Guidance for responders on nation-state attack (2024-01-25)	https://www.microsoft.com/en-us/security/blog/2024/01/25/midnight-blizzard-guidance-for-responders-on-nation-state-attack/	2026-11-03	Confirmed
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71	Mandiant (Google Cloud), APT44: Unearthing Sandworm (2024-04-17)	https://cloud.google.com/blog/topics/threat-intelligence/apt44-unearthing-sandworm	2026-11-03	Confirmed
72	Microsoft On the Issues, Protecting democratic institutions from cyber threats (2024-10-03)	https://blogs.microsoft.com/on-the-issues/2024/10/03/protecting-democratic-institutions-from-cyber-threats/	2026-11-03	Confirmed
73	Microsoft Threat Intelligence, Midnight Blizzard conducts large-scale spear-phishing campaign using RDP files (2024-10-29)	https://www.microsoft.com/en-us/security/blog/2024/10/29/midnight-blizzard-conducts-large-scale-spear-phishing-campaign-using-rdp-files/	2026-11-03	Confirmed
74	Microsoft Threat Intelligence, Frequent freeloader part I: Secret Blizzard compromising Storm-0156 infrastructure for espionage (2024-12-04)	https://www.microsoft.com/en-us/security/blog/2024/12/04/frequent-freeloader-part-i-secret-blizzard-compromising-storm-0156-infrastructure-for-espionage/	2026-11-03	Confirmed
75	Microsoft Threat Intelligence, Storm-2372 conducts device code phishing campaign (2025-02-13, updated 2026-07-31)	https://www.microsoft.com/en-us/security/blog/2025/02/13/storm-2372-conducts-device-code-phishing-campaign/	2026-11-03	Confirmed
76	Microsoft Threat Intelligence, New Russia-affiliated actor Void Blizzard targets critical sectors for espionage (2025-05-27)	https://www.microsoft.com/en-us/security/blog/2025/05/27/new-russia-affiliated-actor-void-blizzard-targets-critical-sectors-for-espionage/	2026-11-03	Confirmed
77	Microsoft Threat Intelligence, Frozen in transit: Secret Blizzard's AiTM campaign against diplomats (2025-07-31)	https://www.microsoft.com/en-us/security/blog/2025/07/31/frozen-in-transit-secret-blizzards-aitm-campaign-against-diplomats/	2026-11-03	Confirmed
78	Google Threat Intelligence Group, New malware attributed to Russia state-sponsored COLDRIVER (2025-10-20)	https://cloud.google.com/blog/topics/threat-intelligence/new-malware-russia-coldriver	2026-11-03	Confirmed
79	ESET Research, Sandworm cyberattack on Poland's power grid in late 2025 (2026-01-23)	https://www.welivesecurity.com/en/eset-research/eset-research-sandworm-cyberattack-poland-power-grid-late-2025/	2026-11-03	Confirmed
80	MITRE ATT&CK, Group G0007 APT28 (last modified 31 July 2026; ATT&CK v20.0)	https://attack.mitre.org/groups/G0007/	2026-11-03	Confirmed
81	MITRE ATT&CK, Group G0016 APT29 (version 6.2, modified 31 July 2026)	https://attack.mitre.org/groups/G0016/	2026-11-03	Confirmed
82	MITRE ATT&CK, Group G0034 Sandworm Team (version 4.2, modified 31 July 2026)	https://attack.mitre.org/groups/G0034/	2026-11-03	Confirmed
83	MITRE ATT&CK, Group G0010 Turla (version 5.1, modified 31 July 2026)	https://attack.mitre.org/groups/G0010/	2026-11-03	Confirmed
84	MITRE ATT&CK, Group G0035 Dragonfly (version 4.0, modified 31 July 2026)	https://attack.mitre.org/groups/G0035/	2026-11-03	Confirmed
85	MITRE ATT&CK, Group G0047 Gamaredon Group (version 3.4, modified 1 October 2026)	https://attack.mitre.org/groups/G0047/	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
86	MITRE ATT&CK, Group G1003 Ember Bear (version 2.1, modified 31 July 2026)	https://attack.mitre.org/groups/G1003/	2026-11-03	Confirmed
87	MITRE ATT&CK, Group G1033 Star Blizzard (version 2.0, created 14 June 2024)	https://attack.mitre.org/groups/G1033/	2026-11-03	Confirmed
88	MITRE ATT&CK, Group G1031 Saint Bear (version 1.0, created 25 May 2024)	https://attack.mitre.org/groups/G1031/	2026-11-03	Confirmed
89	MITRE ATT&CK, Group G1035 Winter Vivern (version 1.0, modified 31 July 2026)	https://attack.mitre.org/groups/G1035/	2026-11-03	Confirmed
90	MITRE ATT&CK, Campaign C0063 2025 Poland Wiper Attacks (version 1.0, created 22 April 2026)	https://attack.mitre.org/campaigns/C0063/	2026-11-03	Confirmed
91	MITRE D3FEND, D3-MFA Multi-factor Authentication	https://d3fend.mitre.org/technique/d3f:Multi-factorAuthentication/	2026-11-03	Confirmed
92	CISA, Known Exploited Vulnerabilities catalog, JSON feed (catalogVersion 2026.11.03, 1,767 entries)	https://www.cisa.gov/sites/default/files/feeds/known_exploited_vulnerabilities.json	2026-11-03	Confirmed
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94	TeamStarWolf library data, `data/attack/campaign_profiles.jsonl` at commit e85b5594	https://github.com/TeamStarWolf/TeamStarWolf/blob/e85b5594301fe061043fe92040135252c97b695d/data/attack/campaign_profiles.jsonl	2026-11-03	Confirmed (data); Reported for TEMP.Veles descriptions
95	TeamStarWolf library data, `data/attack/ics/groups.jsonl` at commit e85b5594	https://github.com/TeamStarWolf/TeamStarWolf/blob/e85b5594301fe061043fe92040135252c97b695d/data/attack/ics/groups.jsonl	2026-11-03	Reported
96	TeamStarWolf library, generated group pages under `mitre/groups/` at commit e85b5594 (page inventory and live-page comparison)	https://github.com/TeamStarWolf/TeamStarWolf/tree/e85b5594301fe061043fe92040135252c97b695d/mitre/groups	2026-11-03	Confirmed
97	TeamStarWolf library, generated page `mitre/campaigns/C0051.md` (APT28 Nearest Neighbor Campaign)	https://github.com/TeamStarWolf/TeamStarWolf/blob/main/mitre/campaigns/C0051.md	2026-11-03	Confirmed
98	TeamStarWolf library, generated page `mitre/campaigns/C0063.md` (2025 Poland Wiper Attacks)	https://github.com/TeamStarWolf/TeamStarWolf/blob/main/mitre/campaigns/C0063.md	2026-11-03	Confirmed
99	TeamStarWolf library, generated page `mitre/groups/G0047.md` (Gamaredon Group)	https://github.com/TeamStarWolf/TeamStarWolf/blob/main/mitre/groups/G0047.md	2026-11-03	Reported
100	TeamStarWolf library data, `data/attack/` at main (technique_profiles, superseded_by, ics and mobile files; the study's technique union)	https://github.com/TeamStarWolf/TeamStarWolf/tree/main/data/attack	2026-11-03	Confirmed

#	Source	URL	Opened	Grade
101	TeamStarWolf library data, `data/attack/group_to_technique.jsonl`	https://github.com/TeamStarWolf/TeamStarWolf/blob/main/data/attack/group_to_technique.jsonl	2026-11-03	Confirmed
102	TeamStarWolf library data, `data/attack/technique_to_d3fend.jsonl`	https://github.com/TeamStarWolf/TeamStarWolf/blob/main/data/attack/technique_to_d3fend.jsonl	2026-11-03	Confirmed
103	Security Affairs, Russia-linked APT Laundry Bear linked to 2024 Dutch police attack (press account)	https://securityaffairs.com/178338/uncategorized/russia-linked-apt-laundry-bear-linked-to-2024-dutch-police-attack.html	2026-11-03	Reported
104	Claims Journal, N.J. appeals court on the war exclusion and the NotPetya attack on Merck (2023-05-04)	https://www.claimsjournal.com/news/east/2023/05/04/316786.htm	2026-11-03	Reported
105	TeamStarWolf library data, `data/weaknesses/cwe.jsonl`	https://github.com/TeamStarWolf/TeamStarWolf/blob/main/data/weaknesses/cwe.jsonl	2026-11-03	Confirmed
106	FBI, Aleksei Sergeyevich Morenets (individual wanted poster for the 2018-10-03 indictment, which states his assignment to Unit 26165), read from an archived copy of this URL captured 2026-03-30	https://www.fbi.gov/wanted/cyber/aleksei-sergeyevich-morenets	2026-11-10	Confirmed
107	FBI, Yuriy Sergeyevich Andrienko (individual wanted poster for the 2020-10-15 indictment, which states his assignment to Unit 74455), read from an archived copy of this URL captured 2026-04-22	https://www.fbi.gov/wanted/cyber/yuriy-sergeyevich-andrienko	2026-11-10	Confirmed

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Related references

- Threat Actors
- Malware Families
- Notable Incidents
- CVE Reference

- Threat Intelligence Reference, Threat Hunting Reference
- Vulnerability Prioritization Reference, Identity Security Reference, Email Security Reference, Network Defense Reference
- Healthcare Security Reference, Ransomware Defense Reference, ICS/OT Security Reference, ICS ATT&CK Technique Atlas
- CWE Reference, CAPEC Reference, D3FEND Reference
- Case studies: All case studies; companion studies on Sandworm operations, SolarWinds and SUNBURST, ICS attacks and advisories, 2024 to 2026, Change Healthcare 2024 and APT41 public record
- Generated MITRE pages: groups APT28 G0007, APT29 G0016, Sandworm Team G0034, Turla G0010, Dragonfly G0035, Gamaredon Group G0047, Ember Bear G1003, Star Blizzard G1033, TEMP.Veles G0088, Saint Bear G1031, Winter Vivern G1035, Wizard Spider G0102 and Indrik Spider G0119; campaigns C0023, C0024, C0025, C0028, C0030, C0032, C0034, C0051 and C0063; software WhisperGate S0689, NotPetya S0368, Uroburos S0022, WellMess S0514, SUNBURST S0559, Spica S1140 and LAMEHUG S9035; techniques Exploit Public-Facing Application T1190, External Remote Services T1133, Password Spraying T1110.003, Valid Accounts T1078, Steal Web Session Cookie T1539, Multi-hop Proxy T1090.003 and Rootkit T1014